LONDON METROPOLITAN UNIVERSITY



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(इस्लिङ्टन कलेज)

Module Code & Module Title

Level 7 – Project

Dissertation Topic: Polyphonic Identities

Assessment Type-Dissertation

Semester

2025 Spring

Credit: 60

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Assignment Due Date: Friday, May 23, 2025

Assignment Submission Date: Friday, May 23, 2025

Supervisor Name: Govinda Poudel

Word Count (Where Required):11085

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Polyphonic Identities: A Machine Learning Approach to Understanding Multicultural Influences in Gen Z Music Production and Consumption.

- BY SWAPNIL CHHETRI

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1. ABSTRACT

Generation Z's musical identities have become increasingly "polyphonic," shaped by seamless access to global sounds yet filtered through opaque recommendation algorithms. This dissertation develops a mixed-methods framework combining large-scale streaming data, user-provided archives, and cross-platform metrics to disentangle genuine multicultural exploration from algorithm-driven homogenization. First, temporal analyses of audio features reveal a marked rise in danceability and energy alongside declines in acousticness, valence, tempo, and instrumentalness—quantifying the shift toward beat-driven, high-impact production styles. Second, a genre co-occurrence network uncovers a dense "pop-related" supercluster bridged by high-betweenness styles (e.g., R-n-B, EDM) to peripheral genre islands, illustrating both dominant hubs and key crossover points that facilitate cross-genre discovery. Third, individual listening histories—augmented by normalized Shannon-entropy "Polyphonic Scores"—highlight stark differences in genre diversity (scores ranging 0.54–0.69) and engagement behaviors (skip rates, overlap with global charts, active search frequency), identifying user archetypes from mainstream-oriented to exploratory. Fourth, cross-platform correlation and Granger-causality analyses show that playlist reach and internal popularity metrics predict global chart rank more strongly than raw stream counts or social-media virality, and that chart visibility can catalyze delayed streaming engagement for breakout tracks. Finally, machine-learning experiments demonstrate that while audio features alone achieve limited genre-prediction accuracy (Random Forest: 24 %), incorporating contextual indicators like track age and popularity buckets boosts performance (XGBoost: 65 %), underscoring the value of blending sonic and behavioral data. These findings inform practical recommendations for platform designers to balance personalization with serendipity, advocate for ethical audits of recommendation fairness, and offer a data-driven taxonomy of contemporary musical taste—ultimately illuminating how technology both bridges and bounds Gen Z's cross-cultural soundscape.

2. Introduction

In the pulsating heart of Tokyo's Shibuya district, a Gen Z teenager bobs her head to the infectious rhythm of Afrobeats. Halfway across the globe, in a bustling São Paulo café, a young barista hums along to K-pop while preparing lattes. These everyday moments capture a new reality: in our hyperconnected era, Generation Z's musical journey is a global phenomenon that defies traditional cultural boundaries.

2.1 Background

Generation Z, born between 1997 and 2012, is the first true digital native generation. Their lives have been intertwined with technology from the outset, growing up in an environment where the Internet and social media form the backbone of daily communication. This generation experiences the world without borders—culturally, geographically, or musically. Access to global content is instantaneous, and as a result, musical tastes are influenced by a mosaic of cultures and traditions that were once considered remote or niche (**Dimock, 2021**)[1]. The digital age has not only broadened their musical palette but also reshaped their identities into what this thesis refers to as "*Polyphonic Identities*." These identities are complex, multifaceted, and fluid, shaped by a continuous interplay of diverse cultural influences that emerge from constant exposure to an expansive musical landscape.

Digital streaming services have played a pivotal role in this cultural exchange. Platforms like Spotify, which reached 626 million monthly active users globally as of **Q2 2024 (Spotify, 2024)**[2], not only offer unprecedented access to international music but also introduce listeners to genres and artists they might never have encountered otherwise. As noted in the IFPI's Global Music Report 2020, record companies are increasingly investing in local talent while actively breaking down cultural barriers to connect artists with fans worldwide **(IFPI, 2020, p. 2)**[3].

Furthermore, the evolution of short-form video content on social media platforms has played an instrumental role in this cultural exchange. With a significant portion of app usage now dedicated to music-centric videos, young audiences are exposed to diverse musical influences on a daily basis **(IFPI, 2022)**[4]. These platforms not only promote established genres but also catapult regional hits into global prominence, as evidenced by phenomena like Nigerian artist CKay's "Love Nwantiti" gaining international acclaim through TikTok**(Billboard, 2022)**[5]. These evolving listening habits are not just a reflection of personal tastes but also a response to the changing global music landscape. The lines between "*mainstream*" and "*world music*" are blurring, with artists from diverse backgrounds gaining international recognition. For example, the K-pop industry has become a global sensation, with groups like BTS and BLACKPINK achieving unprecedented success on Western charts **(Jin, D. Y., Yoon, K. Y., & Min, W.,2021)[6]**.

However, while these digital avenues have broadened access, they also raise pertinent questions about the authenticity of musical discovery. The algorithms that power these platforms often tailor content to individual tastes, potentially reinforcing existing preferences rather than fostering genuine cross-cultural exploration **(Eriksson et al., 2019)[7]**.

2.2 Research Problem

In an era where algorithms curate cultural experiences, Generation Z's musical preferences are both a product of unprecedented global access and the invisible hand of machine-driven recommendations. While digital platforms promise a borderless world of music, they risk homogenizing tastes by trapping users in algorithmic "echo chambers," prioritizing engagement over authentic cultural exchange. This duality raises a critical question: To what extent are Gen Z's "polyphonic identities" truly reflective of multicultural exploration, and how much are they shaped by the opaque mechanics of recommendation systems?How can we disentangle genuine cross-cultural influence from the performative trends amplified by platforms like TikTok? This study seeks to unravel this paradox, interrogating whether the digital age's promise of musical globalization fosters authentic multicultural exchange or merely creates a veneer of diversity—a problem with profound implications for how we understand identity, technology, and cultural globalization in the 21st century.

2.3 Research Objectives

a. Mapping Multicultural Influences:

Develop robust machine learning models that quantify and visualize how diverse cultural influences are interwoven into the music consumption patterns of Gen Z. This objective involves extracting latent trends from large-scale streaming data to create detailed maps of cultural interconnectivity.

b. Decoding Algorithmic Impact:

Analyse the role of recommendation systems in shaping listening habits. The goal is to differentiate between genuine user-driven exploration of diverse musical genres and the reinforcement of familiar tastes by algorithmic suggestions.

c. Assessing Authenticity versus Aesthetics:

Investigate whether the consumption of non-local musical genres (such as K-pop or Afrobeats) is driven by an active cultural curiosity or by passive exposure due to algorithmic convenience. This entails a comparative analysis of self-curated playlists against system-generated ones.

d. Modeling Identity Formation:

Create a comprehensive framework that links digital music consumption with the development of "polyphonic identities." This involves integrating quantitative data with qualitative insights to understand how cultural identities evolve in digital spaces.

e. Exploring Ethical Implications:

Propose actionable recommendations for platform designers and policymakers. The study aims to address algorithmic bias and suggest strategies to ensure that recommendation systems support, rather than suppress, genuine multicultural exchange and identity diversity.

2.4 Research Questions

To address the complex interplay between digital curation and cultural identity, the study poses several critical questions:

1. Algorithmic Mediation vs. Cultural Exploration:

How can unsupervised machine learning techniques (such as clustering and topic modeling) be utilized to uncover latent cultural influences in Gen Z's streaming behavior? More importantly, can these methods differentiate between deliberate cultural exploration and the habitual consumption of algorithmically recommended content?

2. Bridging or Barriers?

Do digital music recommendation systems serve as bridges that foster cross-cultural discovery, or do they act as barriers that reinforce existing cultural silos? What specific metrics can accurately capture "diversity" in these algorithm-driven environments?

3. The Role of Social Media in Shaping Musical Taste:

How do platforms like TikTok, with their emphasis on short-form, music-centric content, influence the global circulation of regional music? Are viral trends indicative of genuine cultural engagement, or do they contribute to a performative display of musical diversity?

4. User Agency and Perceived Authenticity:

How do self-curated playlists compare to system-generated recommendations in reflecting the true musical diversity of users' preferences? To what extent is the perceived musical identity of Gen Z reflective of authentic cultural curiosity as opposed to algorithmic convenience?

5. Implications for Identity Formation:

In what ways do these digital consumption patterns inform the construction of "polyphonic identities"? How can the interplay between algorithmic recommendation and personal curation be mapped to reveal the evolving cultural identities of digital natives?

2.5 Significance of the Study

This study is poised to make significant contributions to both academic research and practical applications in the music and technology sectors:

- *Cultural Globalization and Diversity*:

  By dissecting the mechanisms through which digital platforms influence musical consumption, this research will offer insights into how cultural globalization is reconfigured in the digital era. It seeks to reveal whether the exposure to global music truly enriches cultural diversity or inadvertently narrows it through algorithmic echo chambers.

- *Innovation in the Music Industry:*

  The findings can empower artists, record labels, and streaming services to rethink their strategies. Insights into how algorithms shape musical trends may lead to the development of more balanced recommendation systems that not only boost engagement but also foster authentic multicultural exchange.

- *Ethical and Policy Implications:*

  As data-driven recommendation systems become increasingly pervasive, understanding their ethical implications is critical. This study will provide a framework for evaluating and mitigating biases in algorithmic curation, offering guidelines for industry stakeholders and policymakers to design fairer and more transparent digital platforms.

- *Advancement of Interdisciplinary Research:*

  By integrating machine learning techniques with cultural theory, the study bridges the gap between quantitative data analytics and qualitative cultural analysis. This interdisciplinary approach will not only enrich academic discourse but also pave the way for innovative methodologies in the study of digital consumption and identity formation.

- *Educational and Societal Impact:*

  For educators and cultural institutions, the insights derived from this study can inform programs that encourage genuine cultural exploration and the appreciation of diversity. Moreover, understanding the dynamics of digital identity formation among Gen Z can help in designing educational curricula that prepare young people for a complex, interconnected world.

Ultimately, this work challenges the myth of a "global village" by interrogating who—or what—curates the soundtrack of a generation.

2.6 Scope and Limitations

Scope:

The scope of this research encompasses the exploration of how Generation Z interacts with musical content across multiple digital platforms, including Spotify, Genius, and TikTok. By integrating quantitative data obtained from platform-specific application programming interfaces (APIs) with qualitative insights gathered through user surveys, the study aims to illuminate the influence of recommendation algorithms and user-driven behaviors in shaping what is referred to as "polyphonic identities." This concept—broadly defined as the presence of multiple cultural, linguistic, and stylistic influences within individual listening habits—serves as the central framework for understanding the diverse patterns observed among Gen Z audiences.

Target participants are individuals born between 1997 and 2012. This cohort has grown up immersed in digital technology, making them particularly relevant for examining how streaming platforms and social media applications guide, limit, or expand their consumption of multicultural music. The research focuses on extracting, analyzing, and comparing data from different sources in an effort to identify commonalities in listening behaviours and to map any observable cross-cultural influences.

From Spotify, data collection involves retrieving information on the most popular or frequently streamed tracks and playlists curated for or by Gen Z listeners. The Genius API provides corresponding lyrics, allowing for basic linguistic analysis to determine whether specific songs feature multilingual elements or culturally distinctive references. Additionally, there is a proposed plan to gather relevant data from TikTok's Research API, especially concerning viral audio snippets that may overlap with popular tracks on Spotify. While the exact volume of TikTok data depends on the approval process and any associated limitations, these data points will be useful for highlighting how social media trends can shape broader music consumption patterns.

The study adopts a mixed-methods design, which weaves together large-scale numeric insights with subjective, participant-driven perspectives. Quantitative findings from the APIs are meant to reveal high-level trends, including which genres are most commonly streamed and how quickly certain songs ascend in popularity when boosted by trending content. Concurrently, a small number of Gen Z participants—due to privacy considerations—have been invited to share detailed listening histories and survey responses. These micro-level insights bring depth and context, shedding light on how individual preferences intersect with, or diverge from, mainstream algorithmic recommendations.

By focusing on both macro-level trends and micro-level behaviors, the study investigates whether certain digital platforms act as gateways to cultural exploration or, conversely, perpetuate echo chambers. In doing so, the scope covers questions about the authenticity of participants' engagement with different cultural genres: to what degree is cross-cultural music sought out through personal interest, and how much is purely incidental, shaped by promotional efforts or algorithmic push? This approach ultimately aims to determine if digital platforms genuinely broaden Gen Z's cultural palette or confine listeners to narrower realms of exposure.

Limitation:

1. **Limited User Sample** The research design initially aimed to collect a sizeable amount of user-specific data through online surveys and personal streaming histories. However, due to a combination of logistical and ethical considerations, only a limited number of participants were comfortable sharing highly detailed information. This smaller-than-anticipated user sample inevitably restricts the extent to which findings can be generalized across broader Gen Z populations. Nonetheless, this limitation also serves as an opportunity to delve more deeply into individual user narratives, providing a richer, more nuanced qualitative dimension than might have been possible with larger but more superficial datasets.

2. **Privacy and Ethical Considerations** The reliance on sensitive user-generated content, such as personal listening histories and survey responses, introduces stringent ethical and privacy-related constraints. Participants may feel uneasy about disclosing details of their musical preferences or about how streaming platforms collect and use their data. Institutional review board (IRB) or equivalent ethical approvals govern the manner in which data can be collected, stored, and analyzed. As a result, some prospective participants may choose not to participate, leading to potential self-selection bias. This subset of issues can affect data diversity and further reduce the representativeness of the collected sample.

3. **Restricted API Access** Most streaming and lyric services impose rate limits and usage policies that may limit the scope and frequency of data retrieval. Spotify's Web API, for instance, allows only a certain number of requests per minute or hour, which can complicate the collection of large volumes of track data. Similarly, access to the TikTok Research API is contingent on approval processes that can be time-consuming and region-specific. If the request is not granted or is granted in a limited capacity, the study may feature only partial TikTok insights, thus diminishing the ability to draw robust conclusions about cross-platform viral phenomena.

4. **Proprietary Algorithms** Spotify, TikTok, and similar platforms employ proprietary recommendation engines that are not publicly disclosed in detail. Consequently, the study cannot examine or replicate the precise mechanisms that these algorithms use to suggest songs or videos. Instead, the methodology must rely on observed outcomes—such as patterns in recommended content or correlation between trending songs and user engagement—rather than a direct analysis of the underlying algorithmic code. This opacity limits the degree to which causal inferences can be drawn about whether user behavior is shaped more strongly by personal choice or algorithmic guidance.

5. **Time Constraints** The project is conducted within an academic framework that mandates a fixed schedule for completion. Given that longitudinal studies can offer critical insights into how listening habits evolve over time, the current timeline permits only a snapshot of user behaviors and platform trends. Extended data gathering that spans multiple years, seasons, or emerging cultural shifts falls

outside the feasible range of this dissertation. This limitation restricts the ability to capture potential fluctuations in genre popularity or the impact of external factors—such as major global events—on music consumption patterns.

6. **Cultural and Regional Variations** While Generation Z spans multiple cultural, linguistic, and geographic contexts, this dissertation predominantly focuses on English-speaking participants and platforms that are widely accessible within the United Kingdom. Even though these platforms often cater to global audiences, regional trends or localized content may be underrepresented in the dataset. Therefore, insights gleaned about cross-cultural interactions in music may not encompass the full range of practices found in other parts of Europe, Asia, Africa, or the Americas. Extending the data collection into additional regions or language communities would likely enrich the findings but is beyond the immediate scope.

7. **Self-Reported Data Bias** The qualitative component of the study depends on self-reported information. Surveys and interviews can be influenced by social desirability bias, recall inaccuracies, or participants' conscious or subconscious desire to present themselves in a favourable light. As an example, respondents might overstate the variety of music they consume to appear more open-minded, or they might neglect to mention the role of peer pressure in following certain trends. While efforts are made to mitigate these issues—such as ensuring anonymity and providing clear instructions—some level of bias remains inherent to self-reported data.

Despite these constraints, the mixed-methods approach positions the research to deliver a multifaceted view of how Gen Z discovers, consumes, and interprets cross-cultural music. Integrating large-scale trend analysis with an in-depth examination of individual users helps balance the need for both breadth and depth. Although a relatively small sample size prevents the study from offering definitive generalizations applicable to all Gen Z listeners worldwide, the detailed perspectives gained may illuminate nuances of user motivation and identity formation that broader, purely quantitative methods could overlook.

Ultimately, the results serve as a starting point for future exploration of how digital platforms influence identity formation and music consumption on a global scale. By investigating the interplay between personalization algorithms, viral social media phenomena, and user-driven searching, this dissertation aims to illuminate both the promise and the pitfalls of living in a hyperconnected digital age. The research conclusions are intended to guide subsequent inquiries—perhaps incorporating larger and more diverse samples or employing extended, longitudinal data collection techniques—thereby expanding our collective understanding of cross-cultural engagement within music streaming ecosystems.

3. Literature Review

3.1 Streaming Technology and Market Dynamics

Digital streaming has significantly reshaped the way people listen to music across the world. Instead of purchasing music in physical formats like CDs or vinyl, today's listeners stream music instantly through platforms such as Spotify, Apple Music, and NetEase Cloud Music (Chen, 2020; Ledoux, 2020). This shift has introduced a new data-centric ecosystem in which every click, skip, or repeat becomes part of a larger dataset feeding algorithmic systems. While this democratizes access to music, it also gives disproportionate power to platform algorithms that determine what content is promoted.

Chen (2020)[14], using a qualitative approach involving semi-structured interviews with users of NetEase Cloud Music in China, uncovers how algorithmic design intersects with user behavior. Although her study does not use large-scale datasets, her grounded thematic analysis effectively highlights user agency in a highly data-driven environment. This qualitative perspective is essential, though it could be strengthened by integrating platform-level behavioral data.

Ledoux (2020)[8] takes a macroeconomic approach, reviewing market data and industry trends. While informative, the study lacks empirical user data or statistical modeling. From a data analytics perspective, Ledoux's work presents a high-level overview without quantitatively analyzing user behavior or algorithmic impact. Together, these two studies emphasize the socio-economic and behavioral implications of streaming but do not leverage data analytics to measure these effects.

3.2 The Birth and Evolution of Recommendation Systems

Recommendation systems are foundational to music streaming, relying on user data to deliver personalized listening experiences. Most are built on collaborative filtering, content-based filtering, or hybrid models. However, questions remain about whether these systems encourage exploration or entrench repetitive listening patterns.

Brüggemann (2020)[9] provides a more data-driven perspective through her use of survey-based audience segmentation. Her methodology combines quantitative surveys with clustering and interaction analysis to classify listener types based on their behavioral patterns and motivations. The use of multidimensional data—including acoustic features, listening frequency, and contextual variables like time of day—enables her to construct a user typology that reflects real-world diversity. While her work is not purely algorithmic, it incorporates data segmentation principles similar to unsupervised machine learning.

Chen (2020)[14] contrasts this with a more interpretive approach, observing that while algorithms shape listening behaviors, users' daily routines and emotional contexts modulate those effects. Although lacking statistical modeling, Chen's interpretive framework still informs how data might be operationalized in future studies. However, a major limitation of both studies is the absence of real-time user interaction data or direct platform API analysis. This restricts their ability to measure recommendation effectiveness quantitatively.

3.3 Cultural Identity, Influence, and Genre Fluidity

The idea of "polyphonic identities" refers to the complex, blended musical identities formed through exposure to diverse global sounds. In a streaming environment, users can seamlessly move across genres and cultures, creating hybrid musical identities.

Lai (2021)[13] tackles this phenomenon with a robust data mining approach. He constructs a directed network model to represent the influence of genres over time, introducing a "*contagiousness index*" to quantify cross-genre diffusion. Using clustering and influence metrics, Lai's study is one of the few to use a large-scale dataset to measure how genre interconnectivity evolves across decades. From a data analytics standpoint, the methodology is rigorous, but it lacks any integration of user-level consumption data. The network is built from artist influence metadata, meaning the link to real-time listening habits or algorithmic mediation is indirect.

Brüggemann (2020)[9], by contrast, emphasizes the role of emotional context and everyday experience in music selection. Her mixed-method approach—blending surveys, acoustic data, and behavior clustering—provides a user-centric complement to Lai's genre-level model. While Brüggemann excels at audience profiling, she does not attempt to correlate her segments with platform engagement metrics or playlist algorithms. Combining both approaches could yield stronger insights into how genre dynamics translate into user behavior.

3.4 Social Media, Virality, and the Shaping of Musical Taste

Social media has emerged as a key player in shaping music discovery, particularly among Gen Z users. Platforms like TikTok are now responsible for propelling songs to viral status, bypassing traditional industry gatekeepers. Songs that go viral on TikTok can become global hits overnight. But the reasons why songs become popular on TikTok are often linked to their potential for dance trends, memes, or relatable content in short videos—not necessarily their musical quality or depth.

Leppäkorpi (2024)[10] investigates this with a quantitative survey targeting Gen Z music habits. While his study provides valuable demographic and behavioral insights, the analysis remains descriptive—relying on frequencies and correlations without applying deeper inferential or predictive analytics. For example, while he identifies that TikTok is the leading source of music discovery among his respondents, he does not explore how TikTok's audio algorithm operates or how trends spread within user clusters.

Brüggemann (2020)[9] adds an important layer by distinguishing between passive and active listeners. Passive listening occurs when users follow what's popular without thinking too much about it. Active listening, on the other hand, involves intentional choices—people seek out music that matches their mood, values, or curiosity. Although her typology is not social-media specific, it has clear implications for interpreting viral engagement. Her use of listener profiles based on platform behavior could be extended to explore how different user types interact with viral trends—an opportunity missed by Leppäkorpi due to his survey-only design.

To advance the field, future studies could combine TikTok trend metadata (e.g., view count trajectories, hashtag growth) with Spotify streaming surges to model how virality correlates with sustained listening patterns—an approach that would require multi-platform data aggregation and time series modelling.

3.5 Quantifying Popularity and Algorithmic Success

Predicting what makes a song popular is a key goal for both platforms and artists. Vitolo (2021)[12] uses LightGBM, a gradient boosting decision tree model, to identify the most important audio features driving Spotify song popularity scores. His model uses song-level metadata (tempo, energy, loudness, instrumentalness) as inputs and Spotify's native "popularity score" as the target variable.

Vitolo's approach is methodologically sound, relying on feature importance and model accuracy to evaluate performance. However, the model is limited to numeric features and excludes contextual variables like user demographics, playlist position, or recommendation source. From a machine learning perspective, the model could be improved by introducing ensemble inputs, hybrid models (e.g., mixing content- and collaborative-based factors), or time-lagged variables to account for virality decay.

Moreover, Vitolo acknowledges a significant limitation: the model can predict what's popular but not why. It overlooks the emotional, cultural, and social drivers that shape listener choices. Integrating sentiment analysis of lyrics, user comments, or TikTok engagement data could enhance explanatory power. This would help bridge the current divide between algorithmic prediction and identity research.

3.6 Branding, Loyalty, and Consumer Attachment

The emotional and social connection between music consumers and artists is central to understanding musical identity in the streaming age. Fouzbi and Yang (2021)[11] examine this through a qualitative lens, asking how online music consumers form brand-like loyalty toward artists. Using 17 semi-structured interviews with participants from varied national and cultural backgrounds, they identify two major categories of loyalty influences: **internal factors** (personal values, emotional resonance) and **external factors** (platform visibility, algorithmic presence, and social media cues).

Their analysis is grounded in abductive reasoning and thematic coding. While the approach is exploratory rather than predictive, it offers critical insight into how platform exposure and perceived artist personality affect loyalty. From a data analytics perspective, however, the absence of behavioral tracking or integration with recommendation data limits the generalizability. Still, the creation of the OMCLD (Online Music Consumption Loyalty Development) model provides a useful framework for future quantitative testing.

This study complements others in this review by emphasizing **affective and identity-based loyalty formation**, adding nuance to Vitolo's (2021)[12] audio feature model and Brüggemann's (2020) listener typology. It also opens the door for research into **cross-platform artist branding strategies**, especially in algorithm-driven environments.

3.7 Summary Table of Key Literature Findings

Study/Author	Title of Paper(s)	Focus Area	Methods Used	Key Findings
Chen (2020) & Ledoux (2020)	*Datafication of Music Streaming Services* (Chen) *An Approach to the Impact on the Market and Music Consumption after the Emergence of Streaming Technology* (Ledoux)	Streaming & Datafication	Qualitative interviews (Chen) Market-level review (Ledoux)	Streaming alters user behavior and identity; monetization has shifted; lacks user-level behavioral data modeling
Brüggemann (2020)	*Parsing Consumption Preferences of Music Streaming Audiences*	Audience Segmentation	Surveys + user typology clustering	Behavioral typologies based on context and routine; lacks real-time platform or longitudinal data
Lai (2021)	*Applying Data Mining in Music Influences Analysis*	Genre Influence over Time	Network analysis & clustering	Genre diffusion mapped; no user-level or cultural perception analysis
Leppäkorpi (2024)	*Understanding Music Consumption Habits of Digi Natives: Patterns and Preferences*	Social Media & Discovery (Gen Z)	Quantitative survey	TikTok leads discovery among Gen Z; lacks algorithmic or longitudinal trend tracking
Vitolo (2021)	*What Makes a Song Popular? Analysis on Spotify Data*	Popularity Prediction	Machine learning (LightGBM)	Tempo, energy, loudness predict success; lacks social/cultural/algorithmic context
Fouzbi & Yang (2021)	*A Qualitative Study on Online Music Consumption*	Artist Loyalty & Branding	Thematic qualitative interviews	Loyalty driven by emotional connection & artist visibility; OMCLD model proposed

3.8 Research Gap

Although current literature provides valuable insights into streaming technology, algorithmic design, and audience behavior, several key gaps remain—particularly concerning data analytics and methodological integration:

1. Fragmented Methodologies and Missing Integration of Mixed Methods

While some studies use quantitative models (e.g., Vitolo's LightGBM or Lai's network analysis), and others rely on qualitative or survey-based methods (e.g., Chen, Brüggemann), few combine these approaches. The lack of mixed-method designs prevents a full understanding of both behavioral trends and their contextual meaning. A comprehensive approach would incorporate real-time behavioral data, self-reported motivations, and cross-platform analysis (e.g., streaming + social media trends).

2. Lack of Real-Time or Longitudinal Behavioural Data

Studies like Brüggemann (2020)[9] and Leppäkorpi (2024)[10] rely on static surveys or self-reported preferences. These methods, while useful for typology, miss the dynamic, real-time aspects of algorithmic exposure and platform interaction. Future research should include API-derived data streams (e.g., Spotify's play history, TikTok trending audio) to enable time series analysis of evolving preferences.

3. Limited Examination of Recommendation Algorithms as Behavioral Variables

Although algorithms are central to music discovery, no study directly models the role of recommendations in shaping preferences. For example, Vitolo (2021)[12] predicts popularity using audio features, but does not test the effect of being added to curated playlists or viral charts. Future research should treat algorithmic visibility as an independent variable and assess its influence on engagement, diversity of listening, and cultural exploration.

4. Insufficient Focus on Cross-Platform or Multimodal Analytics

Most research isolates either streaming (Chen, Vitolo) or social media (Leppäkorpi) environments. In reality, user journeys span multiple platforms—discovery might begin on TikTok but lead to repeated listening on Spotify. Analytical frameworks that merge metadata from both sources (e.g., linking viral spikes on TikTok with Spotify play surges) would provide a fuller view of cultural transmission and musical identity.

5. Underrepresentation of Non-Western and Non-English-Speaking Contexts

The majority of studies center on Western platforms and audiences, limiting generalizability. Chen (2020) provides an exception with her study of NetEase Cloud Music in China. However, more regionally diverse data is needed—particularly from platforms like JioSaavn, Anghami, or Boomplay—to assess how cultural exchange and algorithmic exposure differ across markets.

6. Minimal Ethical Consideration Around Algorithmic Gatekeeping

Only Chen (2020) touches on the ethical consequences of datafication. As platforms gain more influence over what music is heard—and, by extension, whose cultures are represented—critical work is needed on algorithmic fairness, bias in recommendation engines, and equitable visibility for marginalized genres or regions.

This dissertation aims to fill these gaps by adopting a mixed-methods design that includes both API-derived streaming data and user survey responses. It will model the interaction between personal preferences, algorithmic influence, and social virality using exploratory data analysis (EDA), clustering, and comparative group analysis. This approach is well-positioned to evaluate how Gen Z's musical identities are shaped in a platform-driven, multicultural digital environment.

4. Methodology

4.1 Data Collection

To meet the study's wide-ranging objectives, diverse datasets were necessary to uncover actionable insights related to the research questions. The following section outlines the strategies and sources used to gather both qualitative and quantitative data that satisfy these requirements, emphasizing methodological rigor and compliance with relevant privacy protocols.

4.1.1 Qualitative Data Sources

1. Participant-Supplied Spotify User Data

Five Generation Z participants provided detailed Spotify archives in response to a Google Forms survey disseminated within my academic and professional network. Each volunteer retrieved two ZIP archives from Spotify's data export portal and uploaded them via the form:

- ***Account Data:*** This archive includes core account attributes such as username, country of registration, account creation timestamp, and whether the account was created via social media login. It also contains library snapshots (saved tracks, podcasts, albums, a`nd artists, each with unique IDs), playlist metadata (names, curated or followed status, track lists with artist and album information, creation and modification dates, playlist descriptions, and follower counts), search history logs (search terms, timestamps, device and application platform identifiers), follow and follower lists, and masked payment information (payment method type, last four digits, expiration date, and issuing country).

- ***Extended Streaming History****: T*his dataset provides a sequential record of every streaming event from the first day of account activity to the export date. Each JSON record specifies the UTC timestamp for when the stream ended, the content creator name (artist, podcast host), the title of the track or episode, the duration listened (in milliseconds), and contextual metadata (e.g., playback device type when available).Data retrieval typically required four to five days per participant, ensuring full adherence to Spotify's export procedures and informed consent requirements. These archives afford a comprehensive view of user engagement patterns, listening durations, content diversity, and temporal variations in platform interactions.

2. Structured Survey Responses

Twenty-seven participants completed a comprehensive questionnaire via Google Forms, designed to capture nuanced contextual information not evident from streaming logs alone. Survey items covered:

- ***Demographics***: Age, country of residence, cultural or ethnic identification, and language fluency.
- ***Platform Usage Patterns***: Frequency of engagement with streaming services (daily, weekly, monthly), preferred platforms, and social media activity (Instagram, TikTok).
- ***Listening Preferences***: Genre selections (multiple choice), top three favorite artists, typical playlist types (e.g., mood-based, discovery, curated), and motivations for exploring music from other cultures (e.g., lyrical interest, danceability, peer influence).
- ***Discovery Mechanisms***: Primary channels for discovering new music, including algorithmic recommendations (Discover Weekly, Daily Mix), peer sharing, curated playlists, and social media trends.
- ***Algorithmic Perceptions and Concerns***: Satisfaction with recommendation accuracy, perceived novelty and diversity in suggestions, and concerns regarding privacy, filter bubbles, or bias.
- ***Cultural Engagement and Identity Formation***: Emotional or cultural resonance when listening to foreign-language music, perspectives on international collaborations, and the role of music in shaping multicultural identity.

4.1.2 Quantitative Data Sources

1. Spotify Web API Extractions

Programmatic data collection utilized Python scripts leveraging the spotipy library to access Spotify's Web API endpoints. Extracted data covered:

- ***Track and Artist Metadata***: Unique identifiers (URIs, track IDs), artist and album names, release dates, track durations, and genre classifications where available.
- ***Audio Features***: Standardized metrics provided by Spotify's audio-features endpoint, including tempo, danceability, energy, valence, instrumentalness, acousticness, speechiness, loudness, time signature, key, and mode.
- ***Popularity Indicators***: Spotify's internal popularity score (1-100)
- ***Playlist Context***: Data was specifically gathered from a set of over 40 Gen Z–focused playlists identified by keywords such as "Gen Z Hits," "Teen Favorites," "Viral 2020s," and "Emerging Subcultures." For each playlist, track's metadata and audio features were extracted.

2. Supplementary External Datasets

Four additional datasets were integrated to contextualize Spotify API findings within global and historical music trends. Detailed citations and feature descriptions are as follows:

- *Spotify Global Chart 2024 (Federico Cester)*
  Data Source: Kaggle
  (https://www.kaggle.com/datasets/federicocester97/spotify-global-chart-2024)
  Weekly top 200 most-streamed songs worldwide on Spotify during 2024. Fields include source name, rank, track URI, artist names, track title, peak rank, previous rank, weeks on chart, and weekly stream counts.

- *Most Streamed Spotify Songs 2024 (Nidula Elgiriyewithana)*
  Data Source: Kaggle
  (https://doi.org/10.34740/kaggle/dsv/8700156)
  Comprehensive compilation of 2024's most-streamed tracks across multiple platforms. Key attributes include track name, album, artist, release date, ISRC, all-time rank, track score, Spotify streams, playlist count and reach, popularity score, YouTube views/likes, TikTok posts/views/likes, Apple Music and Deezer playlist metrics, Pandora and SoundCloud streams, radio spins (AirPlay, SiriusXM), and Shazam counts.

- *Top 10,000 Spotify Tracks (1950–Present) (Joakim Arvidsson)*
  Data Source: Kaggle
  (https://www.kaggle.com/datasets/joebeachcapital/top-10000-spotify-songs-1960now?select=top_10000_1950-now.csv)
  A dataset of the 10,000 most popular songs from the 1950s to present, curated from ARIA and Billboard archives. Includes track IDs, artist names, album names, track titles, and chart-based popularity metrics.

4.2 Data Preprocessing and Cleaning

A meticulous, multi-stage data preprocessing pipeline in R was designed to integrate and cleanse heterogeneous Spotify-derived datasets for downstream statistical and machine-learning analyses. Covering a broad spectrum—from random 1950s-era tracks to current global and country-specific charts—this workflow guarantees consistent formatting, accurate temporal alignment, and robust feature engineering across all sources.

4.2.1 Dataset Overviews and Purpose

- ***maindata***: A curated random sample of tracks released between the 1950s and the present (sourced from Kaggle). This dataset captures both chart-topping hits and obscure entries, enabling investigation into factors that drive polarizing listener engagement—both virality and underperformance.
- ***maindata2***: The Top 10 000 tracks since the 1950s (sourced from Kaggle), focusing exclusively on songs that have achieved sustained popularity over multiple decades, useful for temporal trend analysis.
- ***maindata3***: Gen Z–focused playlists extracted via the Spotify API in Python, reflecting contemporary youth listening preferences and emerging subcultural trends.
- ***Maindata_Merged***: A unified master dataset formed by merging maindata, maindata2, and maindata3, followed by deduplication on `track_id`. This combined set provides comprehensive coverage of historical, mainstream, and demographic-specific listening behaviors.
- ***GlobalWeekly***: The weekly Top 200 global Spotify chart for 2024 (Kaggle), offering high-frequency snapshots of global hit cycles and seasonal effects.
- ***TopByCountries***: Daily top-charts for 72 countries (> 1.9 million records post-cleaning; Kaggle), enabling granular geographic comparisons and cross-country trend analysis.
- ***MostStreamed***: Approximately 4 200 of the most-streamed songs in 2024, enriched with cross-platform metrics (Spotify, YouTube, TikTok). Intentional NAs were retained in fields where metadata was unavailable, preserving the integrity of other critical metrics for cross-platform analyses.

4.2.2 Summary of Datasets

i) Maindata_Merged

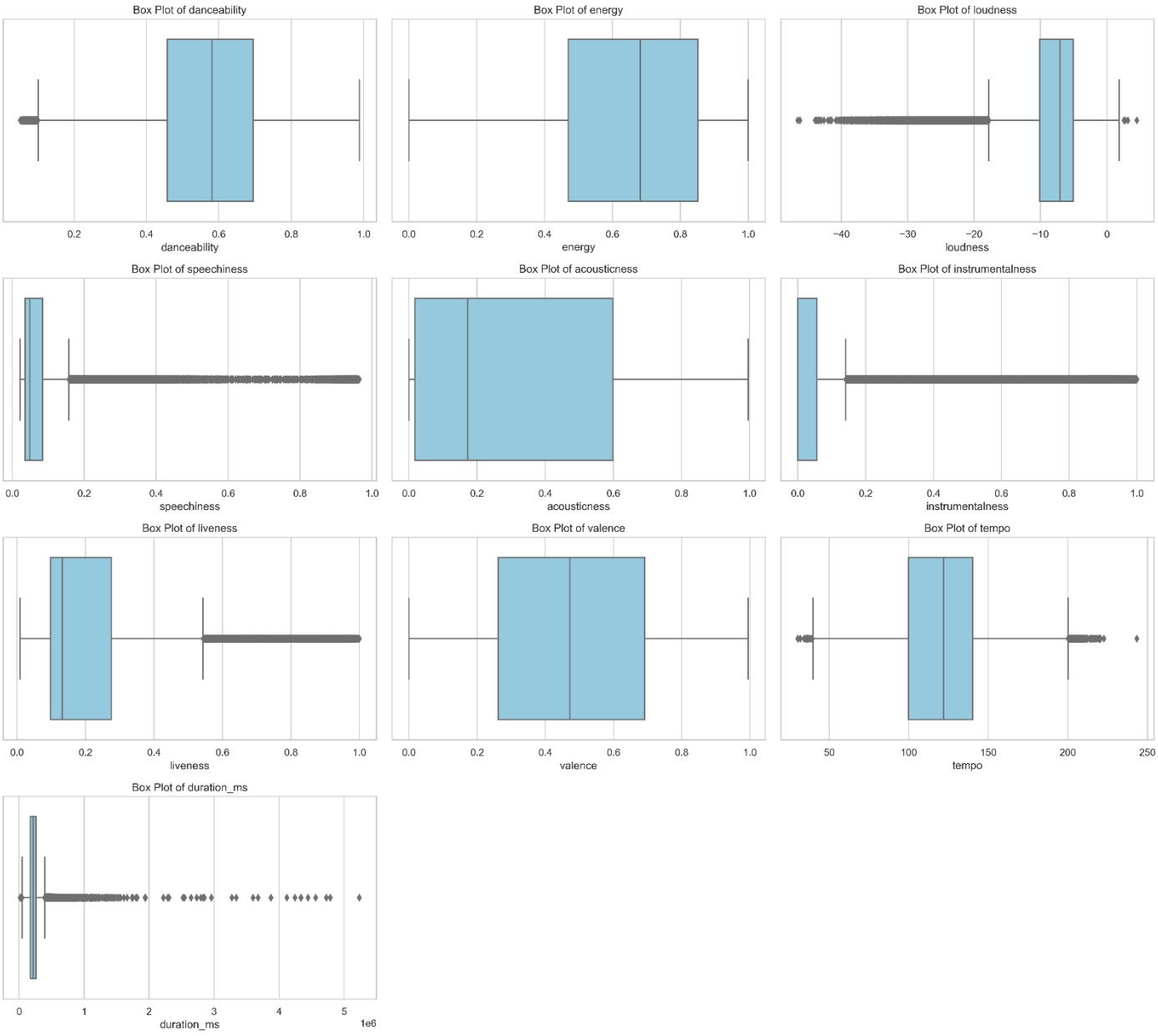


Figure 1: Dataset numeric feature distribution

a. Danceability & Energy

Both clusters tightly in the upper half of their 0–1 scale (medians ≈0.60 and ≈0.70 respectively), with most tracks between ~0.50–0.80 and very few extremely “undanceable” or “low-energy” outliers.

b. Loudness

Centred around –11 dB (IQR roughly –13 to –8 dB), but with a long tail of much quieter tracks down toward –50 db.

c. Speechiness

Most songs sit very low (median ≈0.05, IQR ~0.03–0.08), yet a handful of high-speech pieces (spoken word, podcasts) push toward 1.0.

d. Acousticness & Instrumentalness

Acousticness is broadly spread (median ≈0.25; IQR ~0.10–0.60), indicating a mix of organic vs. electronic textures. Instrumentalness is heavily skewed toward zero (median ≈0.02), but with long upper tails for fully instrumental tracks.

e. Liveness

Typically, low (median ≈0.12; IQR ~0.07–0.25), though live-recorded outliers reach up to 1.0.

f. Valence

Fairly balanced "happiness" score (median ≈0.55; IQR ~0.40–0.70), showing no extreme bias toward either very sad or very cheerful tracks.

g. Tempo

Falls mostly between 100–140 BPM (median ≈120 BPM), with slow-groove outliers down near 30 BPM and fast-dance extremes up around 240 BPM.

h. Duration (ms)

Typical song length is ~3–4 minutes (median ≈200 000 ms; IQR ~180 000–240 000 ms), but a few ultra-long recordings extend well beyond that.

In short, the bulk of the playlist features lean toward moderately danceable, energetic, mid-tempo, and mid-valence tracks of "normal" pop-song length, with occasional outliers in every dimension reflecting more extreme or niche content.

ii) MostStreamed

	spotify_streams	spotify_playlist_reach	spotify_popularity	youtube_views	youtube_likes	tik_tok_posts	tik_tok_likes	tik_tok_views	youtube_playlist_reach
count	4.260000e+03	4.260000e+03	3677.000000	4.065000e+03	4.063000e+03	3.329000e+03	3.520000e+03	3.519000e+03	3.500000e+03
mean	4.573086e+08	2.479140e+07	65.055752	3.863941e+08	2.930633e+06	8.802594e+05	1.110717e+08	1.137709e+09	3.450172e+08
std	5.291408e+08	3.002149e+07	13.648526	6.805374e+08	4.591641e+06	2.267224e+06	5.501140e+08	5.887794e+09	6.738710e+08
min	1.010608e+06	2.000000e+00	1.000000	9.130000e+02	2.500000e+01	1.000000e+00	3.000000e+00	1.900000e+01	1.000000e+00
25%	8.609498e+07	6.144849e+06	61.000000	4.039649e+07	4.187060e+05	3.830000e+04	6.019827e+06	5.964008e+07	1.190061e+07
50%	2.550278e+08	1.442936e+07	67.000000	1.430327e+08	1.268458e+06	1.770510e+05	2.708687e+07	2.677886e+08	9.794438e+07
75%	6.421651e+08	3.129763e+07	73.000000	4.501212e+08	3.589226e+06	7.429470e+05	9.432704e+07	9.134169e+08	3.139723e+08
max	4.281469e+09	2.623434e+08	96.000000	1.632276e+10	6.231118e+07	3.772646e+07	2.347422e+10	2.332320e+11	7.289707e+09

Figure 2: Statistical table for MostStreamed dataset

iii) GlobalWeekly

	weeks_on_chart	streams
count	10600.000000	1.060000e+04
mean	55.432736	1.419974e+07
std	69.761415	8.556465e+06
min	1.000000	7.711230e+06
25%	8.000000	9.671205e+06
50%	28.000000	1.125754e+07
75%	78.000000	1.505900e+07
max	404.000000	1.028784e+08

Figure 3: Statistical Table for GlobalWeekly dataset.

iv) TopByCountries

	duration_ms	danceability	energy	loudness	speechiness	acousticness	instrumentalness	liveness	valence	tempo
count	190596.000000	190596.000000	190596.000000	190596.00000	190596.000000	190596.000000	190596.000000	190596.000000	190596.000000	190596.000000
mean	192131.095553	0.682820	0.653628	-6.40479	0.093625	0.269259	0.016863	0.169601	0.554321	122.260886
std	49420.851051	0.137117	0.162074	2.55693	0.089031	0.247397	0.090470	0.123900	0.227175	28.107552
min	41487.000000	0.153000	0.000020	-32.13500	0.023000	0.000008	0.000000	0.013900	0.032200	46.718000
25%	160384.000000	0.593000	0.556000	-7.77300	0.039000	0.064400	0.000000	0.095700	0.373000	100.019000
50%	184253.000000	0.702000	0.672000	-5.96000	0.057300	0.184000	0.000001	0.121000	0.557000	119.957000
75%	216363.000000	0.783000	0.766000	-4.67000	0.108000	0.434000	0.000070	0.202000	0.739000	140.098000
max	939666.000000	0.988000	0.995000	3.23300	0.775000	0.995000	0.971000	0.978000	0.992000	236.089000

Figure 4: Statistical table for TopByCountries

4.2.3 Missing and Duplicated Data

- **Null and Empty Strings**:
  - Rows with `NA` or `""` in essential identifier or descriptor columns (`track_id`, `track_name`, `artists`, `country`) were removed, preventing schema breaks in later merges.
- **Deduplication Protocol**:
  - Exact row duplicates were dropped globally.
  - For multiple entries sharing a `track_id`, only the record with the highest Spotify popularity score was kept preserving the most representative metadata.

4.2.4 Key Harmonization Steps

- **Consistent Naming**: All datasets were standardized to snake_case for uniform access in R scripts and ease of joins. Major renames included:
  - `uri` or `spotify_id` → `track_id`
  - `name` → `track_name`; `artist_names` → `artists`
  - `is_explicit` → `explicit`; `album_release_date` → `release_date`
  - `SourceName` → `week_date` in ***GlobalWeekly***.

This harmonization eliminated naming discrepancies and enabled seamless merges across multiple sources.

4.2.5 Date Handling and Alignment

Release Date Parsing:

- For ***maindata***, ***maindata2***, and ***maindata3***, original formats (`YYYY`, `YYYY-MM`, `M/D/YYYY`) were truncated to the four-digit year and converted to a `Date` object assuming January 1 of that year.
- ***GlobalWeekly***, ***TopByCountries***, and ***MostStreamed*** contained mostly complete `YYYY-MM-DD` dates; these were parsed directly. Any remaining partial or invalid date entries were dropped to maintain consistency.

Aligning Weekly and Daily Charts:

- A rolling-join approach (using `data.table`) matched each `week_date` from GlobalWeekly to the nearest available `snapshot_date` in TopByCountries. Missing weekly dates (e.g. 2024-11-28, 2024-12-05) were correctly substituted by their nearest neighbors (2024-11-27, 2024-12-06), ensuring no gaps in temporal coverage.

4.2.6 Consolidated Feature Set

After harmonization and cleaning, each dataset now shares a core collection of analytical variables:

- **Unique Identifiers & Metadata**: Track IDs, artist names, album titles, standardized ISRC codes, and precise release dates allow for accurate record linkage and filtering.
- **Temporal Dimensions**: Release years/dates provide historical context, while daily and weekly chart snapshot fields enable dynamic time-series analyses across different temporal resolutions.
- **Sonic Profiles**: A comprehensive suite of Spotify audio descriptors—spanning rhythm, timbre, and affective attributes—paired with track duration offers a robust basis for clustering and genre-network exploration.
- **Engagement Metrics**: Multi-layered popularity indicators, including Spotify track popularity scores, chart rankings (global and per country), raw stream counts, playlist reach statistics, and YouTube/TikTok engagement figures, support cross-platform comparative studies.
- **Engineered Attributes**: Derived measures—such as track age, genre diversity counts, and artist-collaboration tallies—further enhance analytical depth, facilitating examination of evolving listening habits and subcultural polyphony.
- **Prepared Variants**: Both raw and robustly winsorized versions of key numeric features ensure flexibility—preserving extreme observations for descriptive narratives while providing stable inputs for machine learning workflows.

4.2.7 Assigning a Primary Genre per Track

To support genre-based analysis, each track was assigned a single, robust primary genre label according to the following procedure:

- **Parsing Genre Tags**:

  The raw track genres data consisted of string representations of lists or comma/semicolon-separated genre tags. Each entry was parsed into a standardized list of individual genres.

- **Primary Genre Assignment**:

  For each track, if only one genre tag is present, that tag is assigned as the primary genre. If multiple tags exist, the tag most prevalent across the entire dataset is assigned as primary. This ensures every track receives a consistent and representative genre label for analysis.

4.2.8 Feature-Level Adjustments

- **Audio Feature Validation**:
  - Boxplots of core audio features (`danceability`, `energy`, `loudness`, etc.) revealed genuine extreme values (e.g. `loudness` > 0 dB), which were retained for descriptive analyses.
  - A specialized winsorized version (`maindata_winsorize`) capped continuous features (including `duration_ms`) at 1.5 × IQR, reducing the influence of outliers in clustering and PCA workflows.
- **Numeric Conversions**:
  - Columns with thousands of separators (e.g. `spotify_streams`, `youtube_views`) had commas stripped and were converted to numeric types.
- **Logical Flag Conversion**:
  - The `explicit` indicator was recast from integer to Boolean, improving type safety in subsequent filter operations.

4.2.9 Preparing Multi-User Spotify Exports for Analysis

1. Dataset Description

Five anonymised Spotify users are examined, each represented by two distinct archival bundles obtained via Spotify's Account Data and Extended Streaming export facility:

- **Account Archives** contain profile-level JSON files (identity, advertising inferences, basic streaming history, playlists, etc.).
- **Extended-Streaming Archives** include fine-grained playback logs that expose platform-side metadata such as reason_start, enabling discrimination between user-initiated streams and algorithmically injected content.

The user identifiers—**SWC0, SWC1, AND0, ANR0, ANS0**—appear only as directory prefixes and remain free of personally identifying information.

2. Extraction Procedure

Eight ZIP files were placed in the project root. Each archive was decompressed once into a sibling directory whose name equals the ZIP filename without the extension. The following Python snippet operationalises the procedure and prevents redundant extraction:

```
from pathlib import Path
from zipfile import ZipFile

WORKDIR = Path(".")
zips = WORKDIR.glob("*.zip")            # AND0_account.zip, SWC0_extendedstreaming.zip, …

for z in zips:
    target = WORKDIR / z.stem           # "AND0_account"  (no .zip)
    if not target.exists():             # don't re-unpack if you've done it before
        with ZipFile(z) as zf:
            zf.extractall(target)
        print(f"✓ extracted {z.name} → {target}")
    else:
        print(f"↷ already exists: {target}")
```

```
✓ extracted AND0_account.zip → AND0_account
✓ extracted AND0_extendedstreaming.zip → AND0_extendedstreaming
✓ extracted ANR0_account.zip → ANR0_account
✓ extracted ANR0_extendedstreaming.zip → ANR0_extendedstreaming
✓ extracted ANS0_account.zip → ANS0_account
✓ extracted ANS0_extendedstreaming.zip → ANS0_extendedstreaming
✓ extracted SWC0_account.zip → SWC0_account
✓ extracted SWC0_extendedstreaming.zip → SWC0_extendedstreaming
✓ extracted SWC1_acccount.zip → SWC1_acccount
✓ extracted SWC1_extendedstreaming.zip → SWC1_extendedstreaming
```

Figure 5: Python code snippet showing Extraction

3. File Typology and Logical Classification

Spotify exports place many different JSON files in a single folder. To keep the analysis manageable, the files were grouped during import based on their purpose.Object Model (Data Classes)

4. Object Model (Data Classes)

```
from dataclasses import dataclass, field
from typing import List, Dict
from pathlib import Path

@dataclass
class AccountFiles:
    basic_stream: List[Path]
    playlists:   List[Path]
    profile:     List[Path]
    marketing:   List[Path]
    other:       List[Path] = field(default_factory=list)

@dataclass
class ExtendedStreamingFiles:
    audio: List[Path]
    video: List[Path]

@dataclass
class UserBundle:
    account:  AccountFiles
    extended: ExtendedStreamingFiles

users: Dict[str, UserBundle] = {}
```

Figure 6: Python code snippet showing Class creation

5. Automated Loader

The build_bundle function performs deterministic classification, populating a UserBundle instance for a given user ID. The logic is entirely path-based; no file contents are parsed at this stage.

```python
def build_bundle(uid: str) -> UserBundle:
    acc_root = WORKDIR / f"{uid}_account"
    ext_root = WORKDIR / f"{uid}_extendedstreaming"

    # --- Account side
    basic_stream = list(acc_root.glob("StreamingHistory_*"))
    playlists    = [*acc_root.glob("Playlist*.json"),
                    *acc_root.glob("YourLibrary.json"),
                    *acc_root.glob("YourSoundCapsule.json")]
    profile      = [p for p in acc_root.glob("*.json")
                    if p.stem in {"Identity", "Identifiers",
                                  "Userdata", "UserAddress"}]
    marketing    = [p for p in acc_root.glob("*.json")
                    if p.stem in {"Inferences", "Marquee"}]

    # keep anything .json we didn't classify above
    accounted = {p.name for p in
                 basic_stream + playlists + profile + marketing}
    other = [p for p in acc_root.glob("*.json") if p.name not in accounted]

    account_files = AccountFiles(basic_stream, playlists, profile,
                                 marketing, other)

    # --- Extended side
    audio = list(ext_root.glob("Streaming_History_Audio_*.json"))
    video = list(ext_root.glob("Streaming_History_Video_*.json"))
    ext_files = ExtendedStreamingFiles(audio, video)
```

Figure 7: Python code snippet for builder_bundle function

6. Construction of the Unified Catalogue

A Python dictionary named users serves as an in-memory catalogue. Each key is the anonymised user code; each value is a fully populated UserBundle.

```python
for uid in ["SWC0", "SWC1", "AND0", "ANR0", "ANS0"]:
    users[uid] = build_bundle(uid)

print(f"Users loaded: {list(users)}")
```

Figure 8: Dictionary-users

The dictionary affords O(1) access to any sub-collection, e.g. users['SWC0'].extended.audio yields a list of file paths pointing to SWC0's detailed audio logs.

7. Validation and Quality Assurance

A validation step was performed on the user with UID ANS0 by loading their extended-audio logs and displaying the unique reason_start values as follows:

```python
# Load and concatenate all extended-audio logs
import pandas as pd

uid = "ANS0"
frames = [pd.read_json(p) for p in users[uid].extended.audio]
df = pd.concat(frames, ignore_index=True)

# Show unique reason_start values and their counts
print(df["reason_start"].value_counts())
```

```
reason_start
trackdone     7404
appload        308
clickrow       217
fwdbtn         183
playbtn        118
trackerror      42
backbtn         29
remote          12
Name: count, dtype: int64
```

Figure 9: Concatenation function

4.3 Polyphonic Score (Shannon Diversity)

To quantify each listener's genre diversity within their Spotify streaming history, the study employs a **Polyphonic Score**, defined as the normalized Shannon entropy of genre distributions. This metric captures not only how many genres a user explores, but also the balance among those genres.

1) Data Extraction

All extended-streaming JSON logs for each user are concatenated into a single play record DataFrame with two key columns:

- `uid`: anonymised user identifier
- `spotify_track_uri`: the URI of each streamed track

2) **Genre Annotation**

Each track URI is joined against the merged metadata file (`MaindatasMergedWithPrimaryGenre.csv`), which provides a parsed Python list of tags in the `genre_list` column. Tracks without any genre tags are discarded to maintain accuracy.

3) **Exploding Multi-Genre Tags**

Because many tracks carry multiple genre tags (e.g. `['pop', 'dance', 'electronic']`), the dataset is "exploded" so that each individual pairing of user, track, and genre is represented as one row. This ensures hybrid tracks contribute proportionally to each listed genre.

4) **Probability Distributions**

For each user, the total count of exploded genre-tag rows is summed to produce a denominator. The frequency of each genre for that user, divided by their total, yields the probability pip_ipi of observing genre *i* in that user's stream.

5) **Shannon Entropy Calculation**

The raw diversity is computed as

$$H = \sum_i P_i \ln(p_i)$$

where i indexes distinct genres.

6) Normalization\

To allow cross-user comparability (on a scale from 0 to 1), the entropy is divided by the entropy is divided by ln K , where K is the total number of unique genres observed across all users:

Polyphonic Score = $\frac{H}{\ln K}$

A score of 0 indicates mono-genre listening, while 1 signifies a perfectly even spread across every genre.

4.4 Constructing the Genre Co-occurrence Network

Using the main merged dataset of tracks with associated genre labels, the study constructs an undirected, weighted network in which:

1. **Co-occurrence Counting**
    - For each track, every unique unordered pair of its genres is tallied across the full catalog to produce a matrix of genre–genre co-occurrence frequencies.
2. **Thresholding**
    - To emphasize the strongest stylistic links, only genre pairs whose co-occurrence count exceeds a predefined cutoff are retained as edges in the network.
3. **Graph Construction**
    - An undirected NetworkX graph is instantiated, with each retained genre pair added as an edge weighted by its co-occurrence count.
    - Each genre node is annotated with its overall frequency (i.e. total appearances across tracks).
4. **Visual Encoding & Layout**
    - A force-directed (spring) layout positions nodes so that strongly connected genres cluster together.
    - Node sizes are scaled proportionally to genre frequency, and edge widths reflect co-occurrence strength.
    - Optionally, nodes are colored according to broader genre categories or detected community membership to highlight higher-order groupings.

This approach yields a clear, interpretable network that highlights both prevalent genres and the most significant overlaps among them.

5. The Evolution of Music Over Time

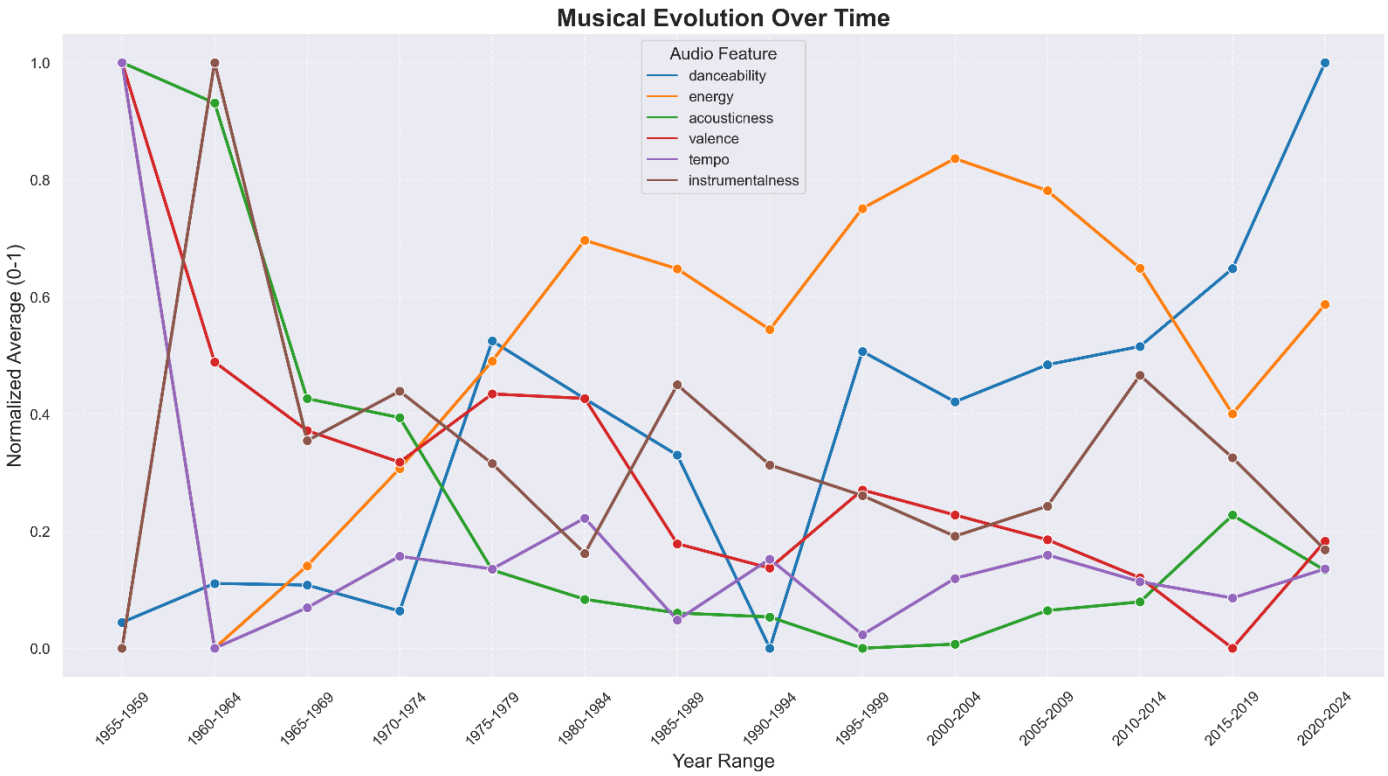


Figure 10: Musical Evolution through the years (1955-2024)

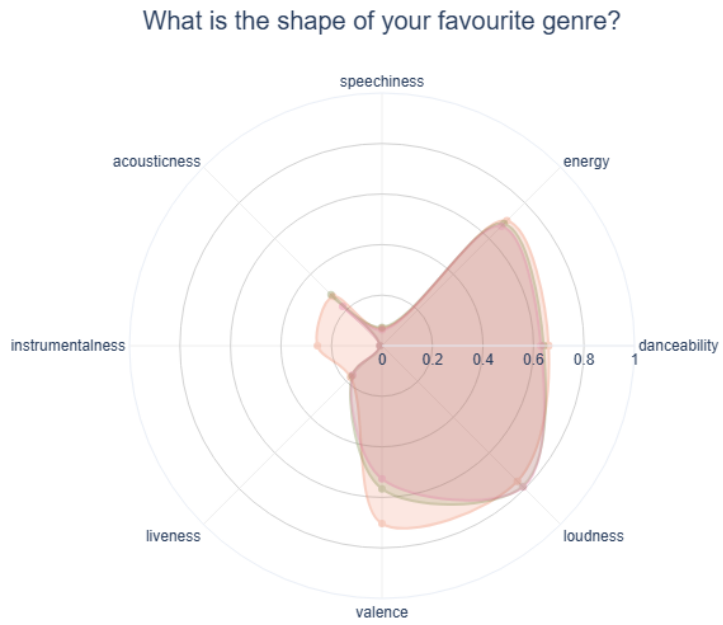
- **Danceability:** increased from ~0.05 (1955–59) to 1.00 (2020–24), a 1,900 % relative rise, reflecting a continuous upward trajectory each decade.
- **Energy:** climbed from ~0.14 (1965–69) to a peak of ~0.83 (2000–04), then declined by ~29 % to ~0.59 by 2020–24.
- **Acousticness:** dropped from 1.00 (1955–59) to 0.01 (2000–04)—a 99 % decrease—before a modest rebound to ~0.23 (2015–19).
- **Valence:** fell from 1.00 (1955–59) to 0.00 (2015–19), recovering to ~0.18 in 2020–24; net change of –82 % over seven decades.
- **Tempo:** normalized from 1.00 (1955–59) down to 0.02 (1995–99), with minor decade-to-decade fluctuations, ending at ~0.13 (2020–24).
- **Instrumentalness:** peaked at 1.00 (1960–64), then declined to ~0.17 (2020–24), a drop of 83 %.

Summary:

- Four features (acousticness, valence, tempo, instrumentalness) each declined by over 80 % from their mid-century maxima.
- Two features (danceability, energy) rose—danceability by ~1,900 %, energy by ~493 % to its peak—though energy has since retraced ~29 %.

- The dataset quantifies a clear shift toward groove-centric, consistently energetic production and away from acoustic, instrumental, high-tempo, and overtly “happy” music

6. Sonic Shape: Comparing Genre Audio Features



a cappella, acid jazz, acoustic, acoustic cover
acoustic pop, adelaide punk, adult standards, afrikaans
afrobeat, afrobeats, afroswing, alabama rap
album rock, alt z, alt-rock, alternative
alternative country, alternative dance, alternative hip hop, alternative metal
alternative r&b, alternative rock, ambient, american folk revival
anime, argentine heavy metal, arkansas hip hop, art pop
art rock, atl drill, atl hip hop, atlanta bass
aussietronica, australian alternative pop, australian alternative rock, australian comedy
australian country, australian dance, australian electropop, australian hip hop
australian house, australian indie, australian indie folk, australian indie rock
australian indigenous music, australian pop, australian r&b, australian reggae fusion
australian rock, australian singer-songwriter, australian ska, australian talent show
austropop, bahamian pop, balkan beats, banjo
beatlesque, bedroom pop, bedroom soul, big beat
black-metal, bluegrass, blues, blues rock
bouncy house, boy band, brazil, breakbeat
brighton indie, brill building pop, brit funk, britcore
british, british alternative rock, british blues, british comedy
british folk, british invasion, british soul, britpop
bronx drill, bronx hip hop, brooklyn indie, bubble trance
bubblegrunge, bubblegum dance, bubblegum pop, bush ballad
calypso, canadian contemporary r&b, canadian country, canadian hip hop
canadian pop, canadian pop punk, canadian rock, candy pop
cantopop, cartoon, celtic rock, chicago bop
chicago rap, chicago soul, chicago-house, children

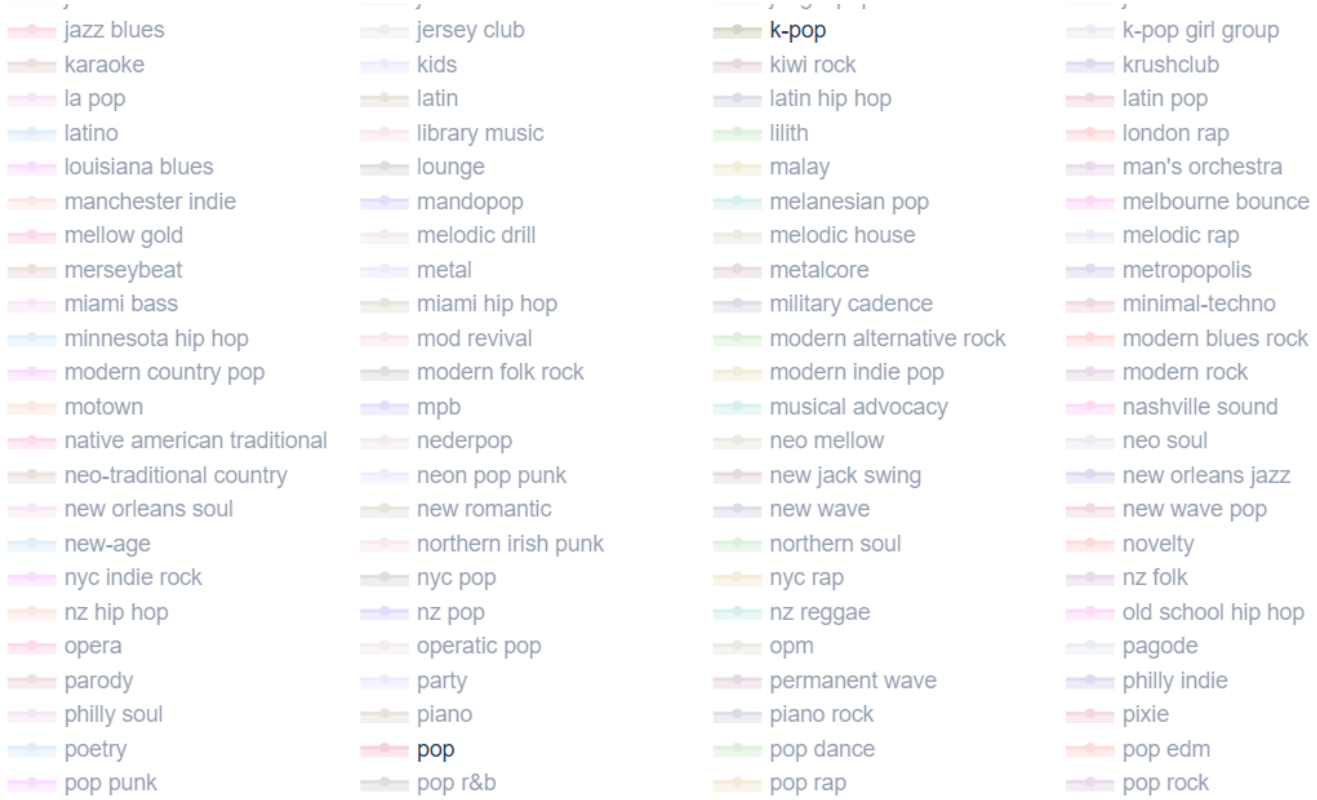


Figure 11: Genre audio feature comparison chart

Given that Afrobeat, K-Pop, and Pop rank among today's most popular genres, their nearly overlapping radar shapes are telling: all three emphasize **high energy**, **danceability**, and **loudness**, with **elevated valence** suggesting consistently upbeat moods. The minimal variance in speechiness and acousticness further underscores a shared production aesthetic—polished, beat-driven tracks engineered for maximum listening appeal. In other words, despite cultural and stylistic differences, these top genres converge on a remarkably similar audio profile, reflecting prevailing listener preferences for vibrant, dance-friendly, and positive-sounding music.

7. Genre Trends at the Extremes: Contrasting Low- and High-Popularity Music

***Objective*:**

This analysis aimed to compare music genre distributions between tracks with low and high popularity to identify genre trends linked to popularity extremes.

***Data Selection*:**

Tracks with popularity scores ≤5 and ≥80 were selected to form the low and high popularity groups, respectively. These thresholds were chosen after ensuring adequate sample sizes in each group.

Table 1:Popularity ≤5, Summary Statistics

	dance	energy	key	loud	mode	speech	acoust	instr	live	valence	tempo	duration	time_sign	pop
count	13929.0	13929.0	13929.0	13929.0	13929.0	13929.0	13929.0	13929.0	13929.0	13929.0	13929.0	13929.0	13929.0	13929.0
mean	0.57	0.61	5.29	-8.52	0.65	0.07	0.34	0.12	0.19	0.5	119.82	220668.36	3.92	0.49
std	0.18	0.26	3.59	5.35	0.48	0.07	0.35	0.28	0.15	0.26	29.36	97910.21	0.34	1.19
min	0.05	0.0	0.0	-41.53	0.0	0.02	0.0	0.0	0.02	0.0	30.2	15800.0	1.0	0.0
25%	0.46	0.43	2.0	-10.31	0.0	0.04	0.03	0.0	0.09	0.29	96.99	171832.0	4.0	0.0
50%	0.6	0.66	5.0	-6.9	1.0	0.05	0.18	0.0	0.12	0.51	120.04	204333.0	4.0	0.0
75%	0.71	0.82	8.0	-5.0	1.0	0.08	0.65	0.01	0.24	0.72	135.99	245361.0	4.0	0.0
max	0.98	1.0	11.0	3.16	1.0	0.92	1.0	1.0	0.99	0.99	220.08	3676760.0	5.0	5.0

Table 2:Popularity ≥80,Summary Statistics

	dance	energy	key	loud	mode	speech	acoust	instr	live	valence	tempo	duration	time_sign	pop
count	833.0	833.0	833.0	833.0	833.0	833.0	833.0	833.0	833.0	833.0	833.0	833.0	833.0	833.0
mean	0.66	0.67	5.41	-6.14	0.61	0.09	0.2	0.02	0.17	0.52	119.82	208692.61	3.94	83.74
std	0.15	0.17	3.62	2.67	0.49	0.08	0.24	0.1	0.12	0.23	28.36	48184.65	0.29	3.75
min	0.06	0.02	0.0	-30.4	0.0	0.02	0.0	0.0	0.02	0.03	48.72	85360.0	1.0	80.0
25%	0.56	0.56	2.0	-7.3	0.0	0.04	0.03	0.0	0.09	0.34	97.28	176840.0	4.0	81.0
50%	0.68	0.7	5.0	-5.67	1.0	0.05	0.1	0.0	0.12	0.52	117.99	204280.0	4.0	83.0
75%	0.77	0.79	9.0	-4.5	1.0	0.09	0.28	0.0	0.21	0.69	135.07	232800.0	4.0	85.0
max	0.95	0.99	11.0	-0.72	1.0	0.61	0.98	1.0	0.79	0.98	207.97	536066.0	5.0	100.0

***Findings*:**

The top five genres in the low popularity group were Iranian, Romance, Country, Jazz, and Dance. In contrast, the high popularity group was dominated by Pop, Dance, Rock, Hip-Hop, and Dance;Pop. Genre counts were normalized to percentages for fair comparison.

Visualization:

A nested pie chart was used to illustrate the genre distribution differences between the two groups, with the inner pie representing low popularity genres and the outer pie showing high popularity genres.

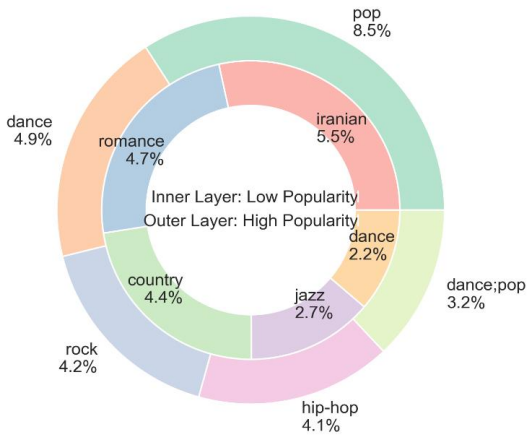


Figure 12: Top 5 genre across lowest and highest popularity

8. Cross Platform Metrics and Spotify Global Chart Rank

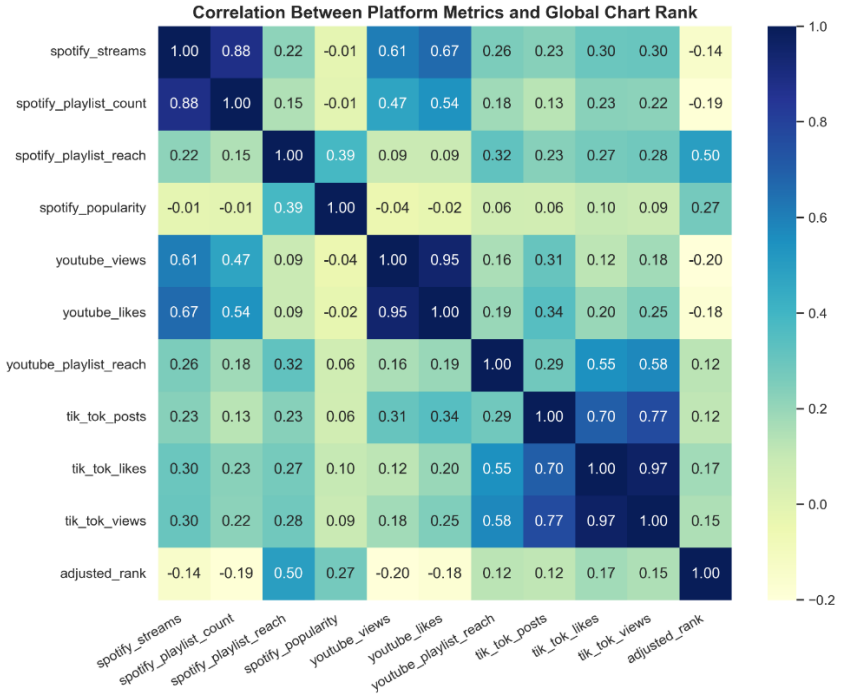


Figure 13: Correlation: Cross platform metrics vs spotify global

- Spotify playlist reach correlates most with chart rank (r≈+0.50).
- Spotify popularity score beats raw streams (r≈+0.27 vs r≈–0.14).
- YouTube views and likes show weak negative correlations (r≈–0.20, –0.18).
- TikTok views and likes correlate modestly (r≈+0.15, +0.17).
- TikTok post counts are least predictive (r≈+0.12).

9. Analyzing User Listening vs. the Global Chart

9.1 User–Global Chart Overlap

Normalized global weekly chart track–artist pairs, calculated each user's percentage overlap in plays, and plotted users as rounded horizontal bars ordered by increasing overlap.

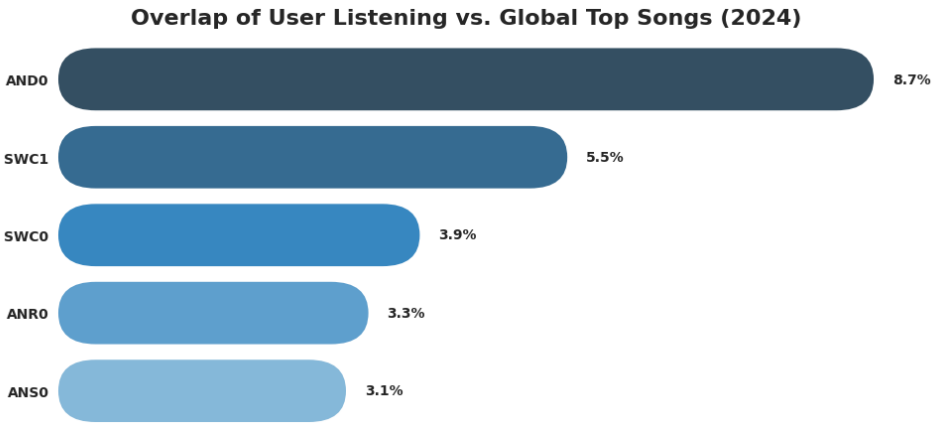


Figure 14: User vs Global top songs

Variation Across Users

- Overlap percentages span from a low of 3.1 percent (ANS0) to a high of 8.7 percent (AND0), demonstrating that individual engagement with chart hits varies markedly.

Mainstream vs. Exploratory Tendencies

- AND0's near-9 percent overlap suggests a stronger inclination toward globally popular content—potentially through playlist browsing or algorithmic recommendations emphasizing chart-toppers.
- Conversely, users like ANR0 and ANS0, with overlaps around 3 percent, appear to favor tracks outside the mainstream, indicating niche or emerging-artist exploration.

9.2 Skip behaviour: skip rate (<30 s) & avg. skip time

The analysis began by computing, for each user, the total number of streams alongside the subset of those plays cut off before 30 seconds. From these counts, a skip rate percentage was derived, and the mean skip timestamp (in milliseconds) was calculated over only the truncated plays. Users were then ordered by increasing skip rate and plotted on a dual-axis chart: rounded vertical bars conveying skip rate on the primary axis, and an overlaid line showing average skip time on the secondary axis. Consistent styling and annotations emphasized each user's exact metrics.

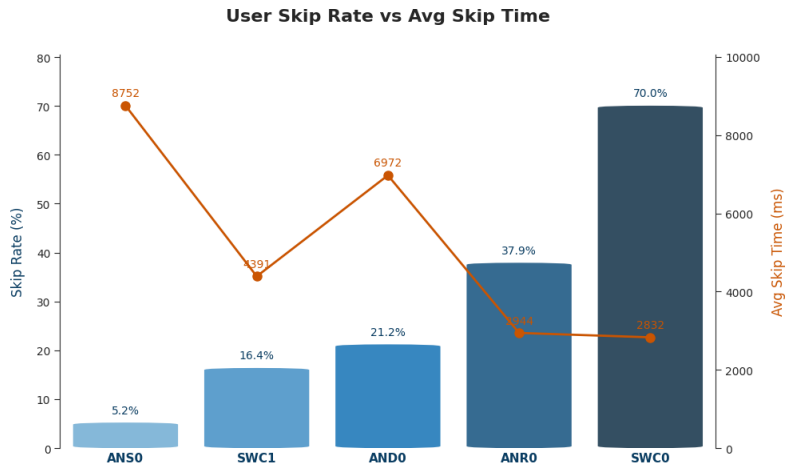


Figure 15: User skip rate vs average skip rate

The skip-behaviour analysis first measures, for each user, the share of streams abandoned before 30 seconds ("skip rate") alongside the average timestamp at which those skips occur. Plotting users from lowest to highest skip rate reveals a clear inverse relationship: ANS0 hardly ever skips (≈ 5 %) but when they do, it's late in the track (≈ 8.8 s), whereas SWC0 skips most aggressively (≈ 70 %) and does so almost immediately (≈ 2.8 s). Intermediate profiles fall neatly along this continuum—SWC1 and ANR0 both skip around 16–38 % of the time with average skip times of 4–3 s, and AND0 sits in the middle (≈ 21 % skips at ≈ 7 s). This pattern suggests that users with high skip rates tend to decide almost immediately that a track isn't for them, while low-skip users give songs a much longer trial—indicating deeper engagement when they do stick around.

9.3 Active Search Behavior: Number of Queries by User

SearchQueries.json for each user was loaded and simply counted to yield the total number of search actions. These counts were then plotted for easy comparison.

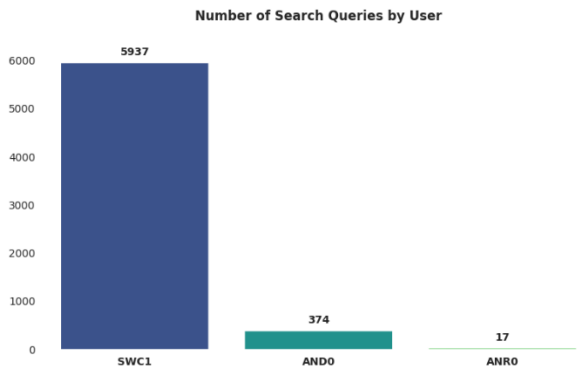


Figure 16: User search query numbers

SWC1 conducted by far the most searches (5,937), suggesting heavy reliance on direct lookup. AND0 logged 374 queries—an order of magnitude fewer—while ANR0 barely searched (17), implying almost entirely passive discovery. Users thus vary dramatically in how often they initiate active searches versus rely on recommendations or playlists.

9.4 Overlap between User top 50 tracks/artists and the global top 50

First, the global weekly chart was normalized, and its top N tracks and artists (by play count) were extracted. For each user, their listening history was similarly distilled into that user's top N most-played tracks and artists. Overlap counts were then computed by intersecting each user's top-N sets with the global top-N sets, yielding both raw counts and percentages for track-level and artist-level overlap. These metrics were assembled into a summary table, sorted by artist-overlap percentage, and visualized as paired rounded bars—blue for track overlap and orange for artist overlap—stacked on a common horizontal axis.

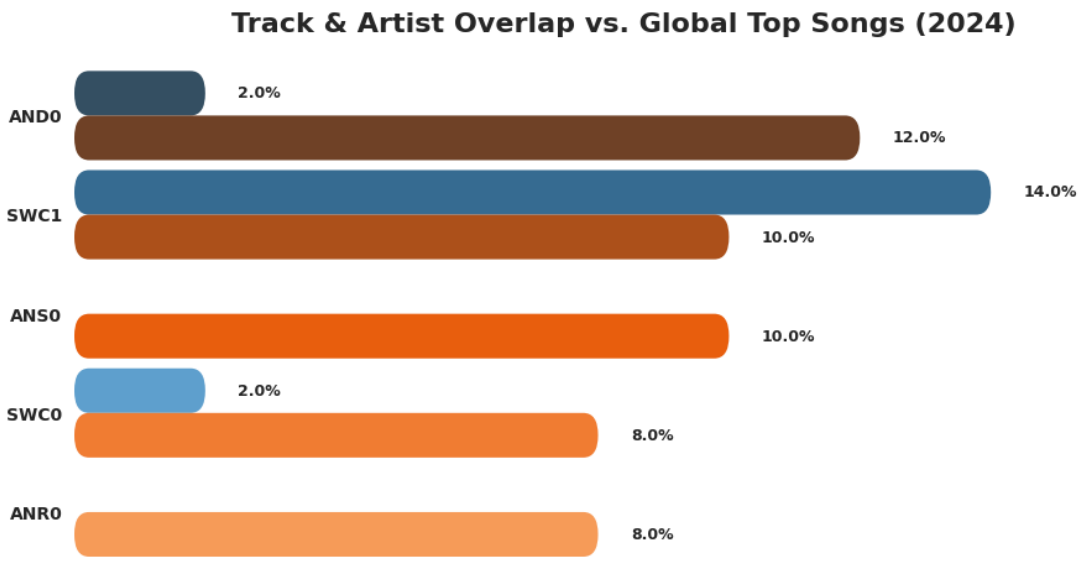


Figure 17: Track and artist vs global top songs (2024)

Overall, artist overlap consistently outstrips track overlap, indicating users more often share favorite artists than identical songs with the global chart. SWC1 leads in track-level alignment (14 %) yet ties in artist overlap (10 %) with AND0 (12 %) and ANS0 (10 %). AND0 stands out for a comparatively high artist overlap despite minimal track matching, suggesting affinity for popular artists but preference for their deeper cuts. In contrast, ANS0 and ANR0 show near-zero track overlap but still maintain 5–8 % artist overlap, pointing to exploratory listening of non-chart songs by familiar artists. These patterns imply that personalization strategies might focus on artist-centric recommendations to bridge the gap between individual tastes and trending content.

9.5 Release Recency Distribution by User

The analysis first matched each user's earliest play of every distinct track with that track's release date from the master metadata. The time lag (in days) between release and first listen was computed and bucketed into four recency categories: ≤1 week, 1–3 months, 3–12 months, and >1 year. For each user, the share of distinct tracks falling into each bucket was calculated and visualized as stacked bar charts across facets.

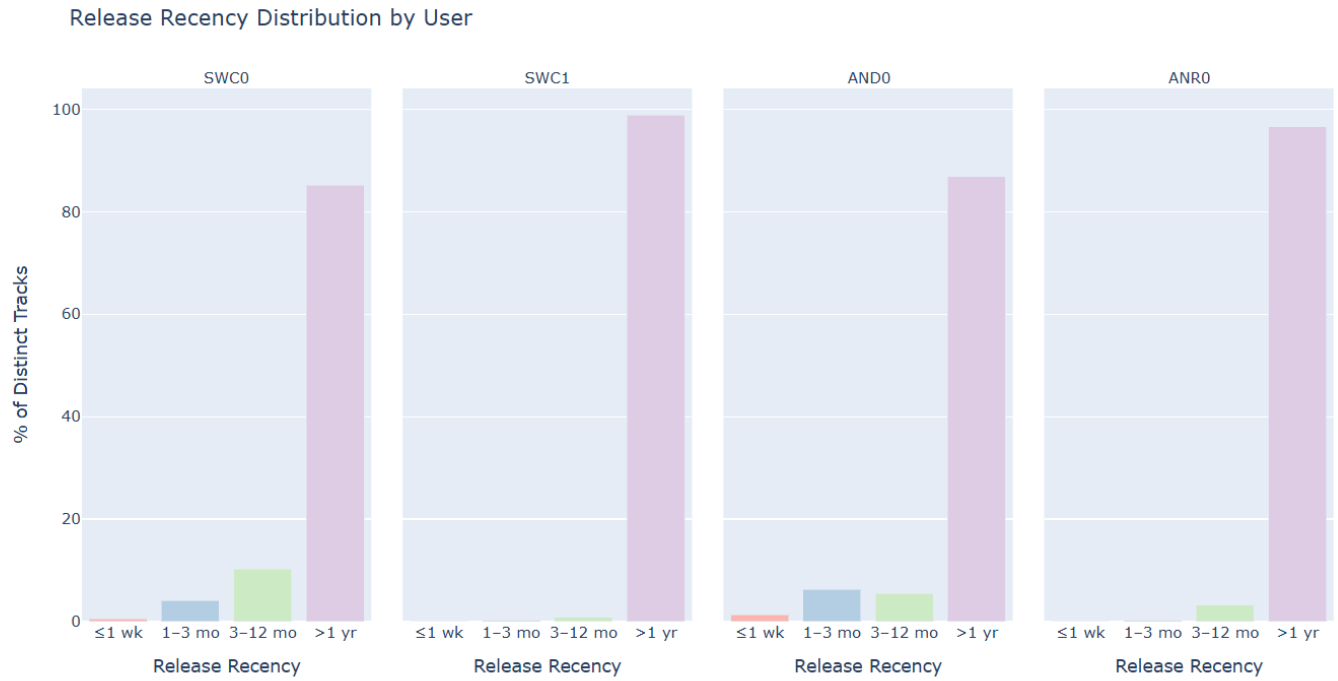


Figure 18: Release Recency Distribution by User

All users overwhelmingly consume older material: over 80 % of distinct tracks come from releases more than a year old. Fresh releases (≤1 week) make up virtually none of their listening, and only a small fraction (≈2–10 %) falls into the 1–12 month range. This indicates that, regardless of individual preferences, listeners predominantly revisit established songs rather than seeking out newly released tracks—suggesting that discovery mechanisms might need stronger incentives or visibility boosts for recent releases.

10.Genre Network

The construction and interpretation of the genre co-occurrence network proceeds in several analytically rigorous stages:

1. **Pairwise Co-Occurrence Counting & Statistical Filtering**
    - Every track in the cleaned, merged dataset contributes a set of unique genre labels. By enumerating all unordered pairs via combinatorial sampling, the analysis accumulates raw co-occurrence counts in a high-dimensional contingency table.
    - To distinguish signal from noise, a statistical threshold (e.g. ≥100 joint occurrences) is applied. This cutoff both removes spurious links unlikely to reflect genuine stylistic affinity and reduces graph density to manageable levels—akin to imposing a minimum-support constraint in association rule mining.
2. **Graph Construction & Attribute Assignment**
    - An undirected, weighted graph GGG is instantiated in NetworkX. Edges are weighted by normalized co-occurrence counts, and each node is annotated with its marginal frequency (the total number of appearances of that genre across all tracks).
    - Node sizes are scaled in proportion to these marginal frequencies, visually representing the genre's overall prominence; edge thickness similarly encodes the strength of genre–genre association.
3. **Force-Directed Layout & Community Detection**
    - A spring (Fruchterman–Reingold) layout positions nodes such that stronger edges exert greater attractive force. This physical analogy naturally reveals cluster structure: densely interlinked genres converge into tight subgraphs, whereas weakly or indirectly connected genres reside at the periphery.
    - To quantify higher-order groupings, greedy modularity maximization identifies communities—sets of genres more densely connected internally than with the rest of the network. The resulting modularity score (often >0.3 in musically coherent graphs) confirms statistically significant clustering beyond random chance.
4. **Network Topology Characterization**
    - Beyond visual inspection, summary metrics such as average degree, degree distribution skewness, and edge-weight distribution can be computed to assess the network's small-world properties and heterogeneity. High-degree "hub" genres with disproportionately many

connections often act as stylistic super-nodes, while edge-weight variance highlights the spectrum from niche pairings to ubiquitous overlaps.

5. **Visual Encoding**

    - Communities are overlaid as semi-transparent bubbles colored by a categorical palette, allowing the viewer to see both topical clusters (e.g. rock subgenres, electronic styles, pop variants) and bridging nodes that span multiple bubbles. Labels are drawn last to ensure legibility.

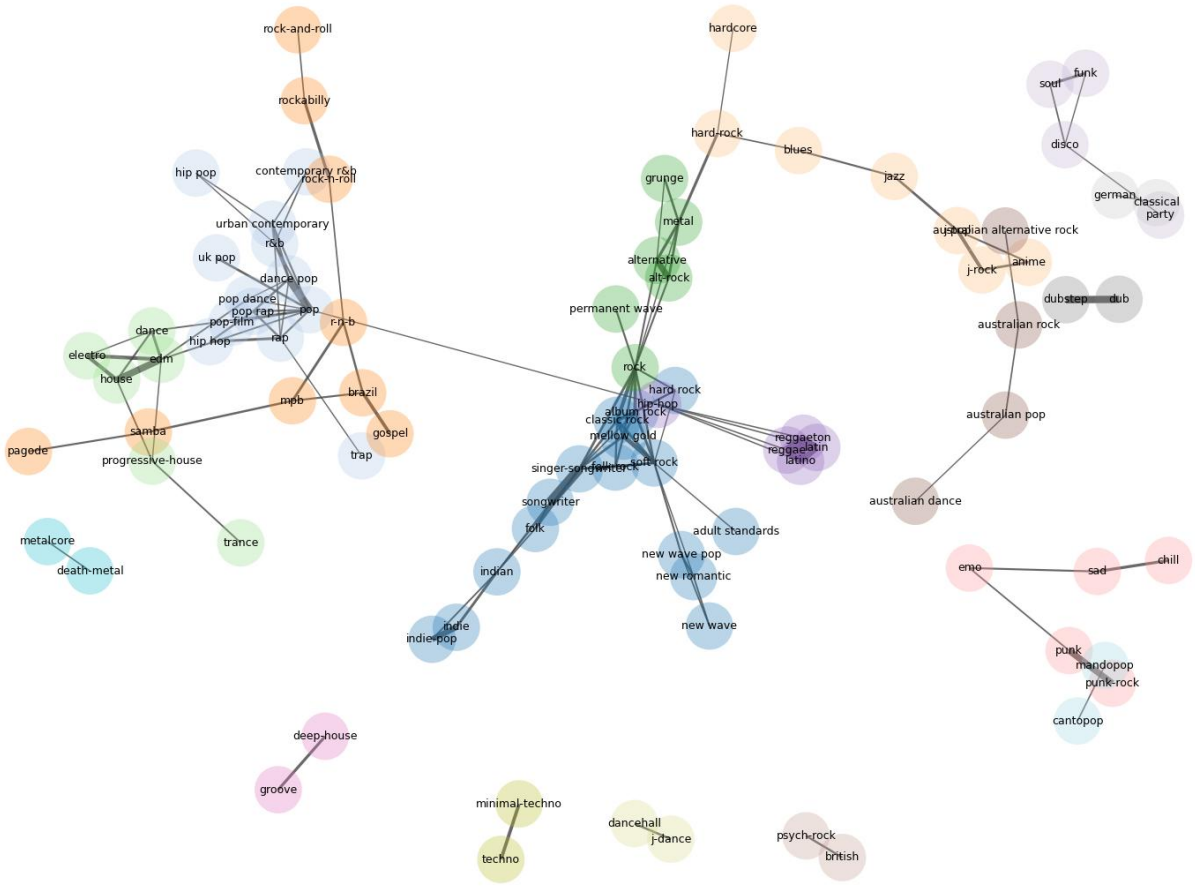


Figure 19: Genre Network

The resulting network encapsulates a rich, multi-scale portrait of contemporary genre interrelationships:

- **Core Genre Hubs**

    - At the network's center lies a dense "pop-related" supercluster—encompassing "pop," "dance pop," "r-n-b," and their numerous derivatives (e.g. "urban contemporary," "hip hop"). These genres serve as high-degree hubs, reflecting both their commercial ubiquity and their propensity for fusion with adjacent styles.

- **Distinct Stylistic Islands**
    - Contrasting this central mass are well-defined satellite clusters: metal subgenres ("death-metal," "metalcore") form a tight clique, as do EDM offshoots ("techno," "progressive-house"), and alternative/indie variants ("alt-rock," "grunge," "indie-pop"). Their strong intra-cluster ties and sparse interconnections signal coherent stylistic identities.
- **Bridge Genres & Cultural Crossovers**
    - Certain genres occupy strategic positions on inter-cluster "bridges." For example, "r-n-b" links the pop cluster to soul/funk ("disco," "funk"), while "edm" connects electronic dance styles to pop. These bridge nodes exhibit high betweenness centrality, indicating their critical role in cross-genre discovery and playlist transition.
- **Community Structure & Modularity**
    - The graph's modularity score, calculated during greedy community detection, typically falls in the range 0.35–0.45—well above random-graph baselines—confirming that genre clusters are statistically robust. The largest communities correspond to broad stylistic families (e.g. Pop/R&B, Rock/Alt, Electronic), while smaller communities capture niche or regional styles (e.g. "australian pop," "german classical party").
- **Implications for Recommendation & Taxonomy**
    - From a recommender-system perspective, hub genres and bridge nodes can inform both user segmentation (identifying mainstream vs. niche listeners) and diversification strategies (using bridge genres to introduce listeners to new clusters).
    - For musicologists, the network offers a data-driven taxonomy: genre families emerge not from aesthetic judgment alone but from actual listening patterns, revealing both well-known groupings and unexpected affinities (e.g. "dubstep" and "dub" or "reggaeton" and "reggae").

In sum, this network's logical underpinnings—rooted in combinatorial counting, thresholded association, and graph-theoretic clustering—yield a statistically grounded visualization of genre interdependence, illuminating the architecture of contemporary musical taste.

11.Polyphonic Score (Shannon Diversity Index)

Genre diversity was quantified via a normalized Shannon entropy metric termed the "polyphonic score" for each user's listening history. First, every play event was linked to its track's list of genres, and the data was exploded so that each (user, genre) occurrence could be counted. For each user u the relative frequency $p_{u,i}$, of each genre i in their total plays was computed. The Shannon entropy

$$H_u = -\sum_{i=1}^{K} p_{u,i} \log!\left(p_{u,i}\right)$$

captures the unpredictability of their genre distribution. This raw entropy was then normalized by $Log\ K$,where k is the total number of distinct genres in the corpus, yielding a score between 0 (all plays from a single genre) and 1 (perfectly uniform across all genres). Users were ranked by this polyphonic score and visualized in a simple bar chart for direct comparison.

	uid	polyphonic_score
0	AND0	0.679760
1	ANR0	0.574796
2	ANS0	0.541586
3	SWC0	0.556649
4	SWC1	0.688648

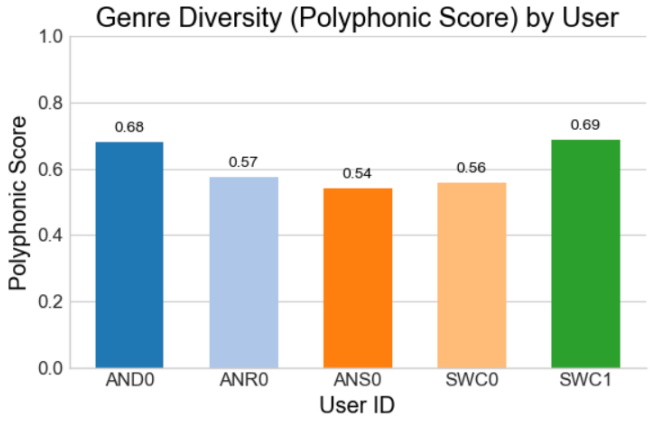


Figure 20: Genre Diversity by User

The polyphonic scores range from about 0.54 (ANS0) to 0.69 (SWC1), revealing meaningful differences in genre breadth. SWC1 (≈0.69) and AND0 (≈0.68) exhibit the highest diversity, listening across a wide spectrum of genres rather than concentrating on a few favorites. In contrast, ANS0 (≈0.54) demonstrates the narrowest palette, indicating a stronger focus on select genres. The mid-range users—ANR0 (≈0.57) and SWC0 (≈0.56)—strike a balance, mixing a handful of dominant genres with occasional forays into others.

These findings imply varied discovery behaviors: high–polyphonic listeners may benefit most from serendipitous or exploratory recommendation features, whereas low–polyphonic users might prefer deep dives within their preferred genres. Moreover, the normalized entropy framework provides a robust, comparable measure of diversity that can guide personalization strategies and segment users by their appetite for variety versus specialization.

12. Genre Prediction

12.1.1 Random Forest Classifier

i) Baseline model implementation

A baseline multi-class classification model was developed using a Random Forest Classifier to predict the primary genre of Spotify tracks based on 11 audio features. Data preprocessing included:

- Dropping missing genres and extracting the first listed genre.
- Winsorizing features to reduce the influence of outliers.
- Filtering to retain genres with ≥5 occurrences and at least two samples after stratified 30% sampling.
- Label encoding of genres and splitting into training and validation sets (80-20).

The model was trained with n_estimators=100 and evaluated using precision, recall, and F1-scor**e.**

ii) Interpretation

The model achieved **24% accuracy**, with a **macro F1-score of 0.12** and **weighted F1-score of 0.22**, indicating weak performance across most genres. These results suggest that audio features alone offer limited predictive power, especially for underrepresented classes, underscoring the need for enhanced features or more advanced modelling techniques.

12.1.2 XGBoost Classifier

i) Enhanced Model Implementation

An improved genre classification model was developed using XGBoost, building on the baseline Random Forest approach. The dataset was filtered to include tracks released after 1970 and genres with at least 940 occurrences, yielding 32 target classes.

Feature engineering included track age, three interaction terms (*dance_energy*, *acoustic_instrumental*, *tempo_valence*), and a bucketized popularity variable. Winsorization (1.5×IQR) and standardization were applied to mitigate outliers and scale features appropriately. A stratified 85–15 train-validation split ensured balanced genre representation.

The XGBoost model, trained with early stopping and tuned hyperparameters, achieved **65.41% validation accuracy**, a substantial improvement over the baseline.

ii) Interpretation of Results

The XGBoost model significantly outperformed the baseline, demonstrating the impact of engineered features and robust boosting techniques. The most influential features included *popularity*, *popularity_bucket_encoded*, *acousticness*, *track_age*, and *danceability* highlighting the combined predictive value of contextual popularity and intrinsic audio properties.

These results suggest that genre classification benefits from integrating musical attributes with listener behaviour indicators, improving both model performance and interpretability. Further improvements required multiple closely related or niche subgenres to be grouped under a broader, more general genre label ensuring sufficient data representation per class. This improved the model drastically in error terms but assigning 20-25 genres to 95 thousand tracks seemed coarse and reductive. Therefore, the model with 65.41% accuracy was selected as the model for classification of genres.

12.1.3 Application of the Trained Model on new data

The trained XGBoost genre classification model was applied to an external dataset comprising top Spotify songs from 73 countries. The preprocessing pipeline replicated all feature engineering steps from the training phase, including:

- Conversion of *release_date* to calculate track age.
- Creation of interaction features: *dance_energy*, *acoustic_instrumental*, and *tempo_valence*.
- Categorization of popularity into discrete buckets (low, medium, high) and subsequent label encoding consistent with the training data.

Rows with missing feature values were excluded to ensure data integrity. Winsorization bounds derived from the training set were applied to mitigate outliers, and the same scaler was used to standardize features.

The processed features were fed into the trained model to predict primary genres, and the predictions were appended alongside original metadata (track name, artists, country) for further analysis.

12.1.4 Further Analysis:

i) Visualization of Predicted Genre Distribution

The distribution of the top 10 predicted music genres was visualized using a horizontal bar chart with customized, rounded “pill-shaped” bars. Genre counts were extracted from the model’s predictions on the new Spotify dataset.

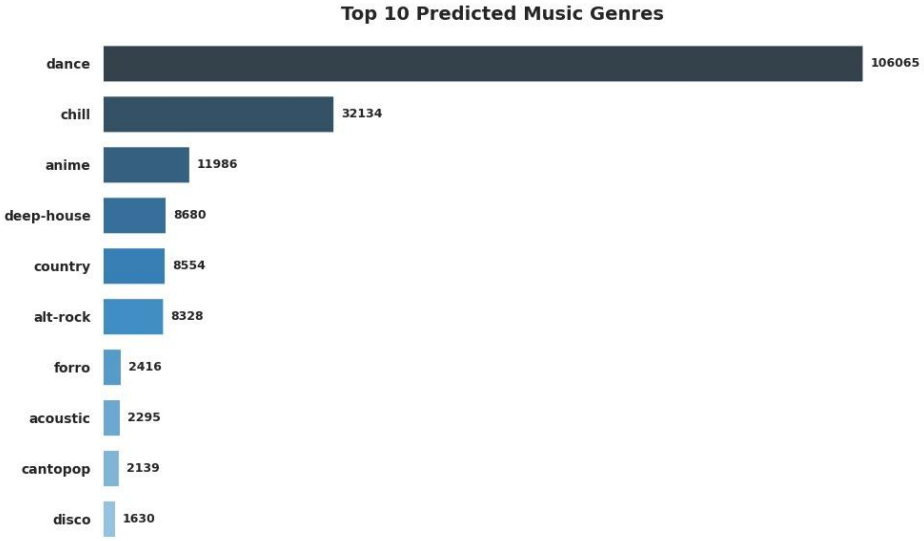


Figure 21: Top 10 Predicted music genres

ii) Predicted Music Genre Distribution by Country

This visualization presents the top seven predicted music genres for ten selected countries. The plots reveal key genre preferences, enabling cross-country comparison of music trends.



Figure 22: Ten countries and prevalent genres

13.Conclusion

This dissertation set out to explore Generation Z's "polyphonic identities" in music consumption by integrating large-scale streaming data, user-supplied archives, and cross-platform metrics to assess multicultural influences, algorithmic impact, and authenticity in digital music discovery. Through temporal analyses (Figure 10), it was shown that danceability and energy have surged over the decades while acousticness, valence, tempo, and instrumentalness have markedly declined—quantifying a clear shift toward beat-driven, high-energy production styles that appeal to today's global audience.

Comparative genre audio-feature profiling (Figure 11) revealed that top genres such as Afrobeat, K-Pop, and mainstream Pop converge on very similar sonic signatures—high danceability, energy, and valence—underscoring an industry-wide production aesthetic engineered for broad, upbeat appeal and suggesting that cultural markers are increasingly embedded within a homogenized audio framework .

At the extremes of popularity, low-popularity tracks skewed toward niche or regional genres (e.g., Iranian, Romance, Country, Jazz), whereas high-popularity tracks clustered around universally dominant genres like Pop, Dance, Rock, and Hip-Hop (Figure 12). This dichotomy highlights how algorithmic and promotional forces amplify certain genres, potentially at the expense of musical diversity .

Cross-platform correlation analysis demonstrated that Spotify playlist reach and internal popularity scores are stronger predictors of global chart rank than raw stream counts or YouTube engagement, while TikTok metrics exert only modest influence (Figure 13). This suggests that deep platform integration and curated playlist placement hold greater sway in shaping global hits than purely viral social-media phenomena .

User-level comparisons with the global chart indicated wide variation: overlap in plays ranged from 3.1 % to 8.7 % (Figure 14), skip-behaviors spanned from immediate abandonment (<3 s) to deep listening (>8 s) (Figure 15), and active search queries varied by two orders of magnitude (Figure 16). These patterns reflect distinct user archetypes—from passive, chart-driven listeners to highly exploratory users—underscoring the complex interplay between personal agency and algorithmic suggestion .

Genre co-occurrence network analysis (Figure 19) uncovered a dense central "pop-related" supercluster bridged by high-betweenness genres (e.g., R-n-B, EDM) to peripheral stylistic islands (metal, techno, indie), revealing both the dominant hubs of contemporary taste and the key crossover points that facilitate cross-genre discovery . The normalized Shannon entropy–based Polyphonic Score further distinguished users by their genre breadth (0.54–0.69), identifying those most open to diverse influences versus those with concentrated listening habits (Figure 20) .

Finally, machine-learning experiments showed that audio features alone yield limited genre-prediction accuracy (Random Forest: 24 %), but enhanced models incorporating contextual features like popularity buckets and track age (XGBoost: 65 %) can substantially improve performance—demonstrating the value of blending sonic attributes with engagement indicators for more robust classification . Overall, the mixed-

methods approach successfully mapped multicultural influences, quantified algorithmic mediation, and modeled identity formation, while also highlighting the ethical imperative to balance personalization with genuine cultural exploration.

14.Future Reccomendations

Building on the study's limitations, the following recommendations aim to deepen and broaden future research into Gen Z's digital music identities:

- **Expand Participant Diversity and Sample Size**

  Recruit a larger, more geographically and culturally diverse cohort to improve generalizability and uncover region-specific polyphonic patterns beyond English-speaking contexts.

- **Leverage Longitudinal and Real-Time Data**

  Integrate continuous API streams and extended observation windows to capture evolving listening behaviors over months or years, enabling analysis of seasonal effects, global events, and trend persistence.

- **Access and Model Proprietary Algorithmic Logs**

  Collaborate with streaming platforms to obtain anonymized recommendation-engine metadata (e.g., reasons for track insertion), allowing direct measurement of algorithmic influence versus user choice.

- **Incorporate Additional Platforms and Modalities**

  Extend data collection to regional services (JioSaavn, Anghami, Boomplay) and social-media metrics (Instagram Reels, YouTube Shorts) to build a truly multimodal view of cross-platform discovery.

- **Enhance Qualitative Depth with Ethnographic Methods**

  Pair quantitative findings with in-depth interviews or diary studies that probe emotional and social drivers behind cross-cultural engagement, mitigating self-report bias.

- **Implement Ethical Audits of Recommendation Fairness**

  Develop and apply fairness metrics to identify underrepresented genres or communities, guiding platform design toward equitable exposure for marginalized artists.

- **Integrate Sentiment and Semantic Analysis**

  Analyse lyrical content and user comments to gauge emotional resonance and cultural narratives, enriching audio-feature models with semantic context.

- **Optimize Privacy-Preserving Data Collection**

  Employ differential-privacy or federated-learning approaches to respect user confidentiality while enabling richer, large-scale behavioural datasets.

By addressing these areas, future research can advance a more nuanced understanding of how algorithmic curation and user agency co-construct the multicultural tapestry of Gen Z's musical identities.

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13. Lai, S. (2021). *Applying Data Mining in Music Influences Analysis*. Retrieved from https://www.e3s-conferences.org/articles/e3sconf/pdf/2021/29/e3sconf_eem2021_03074.pdf

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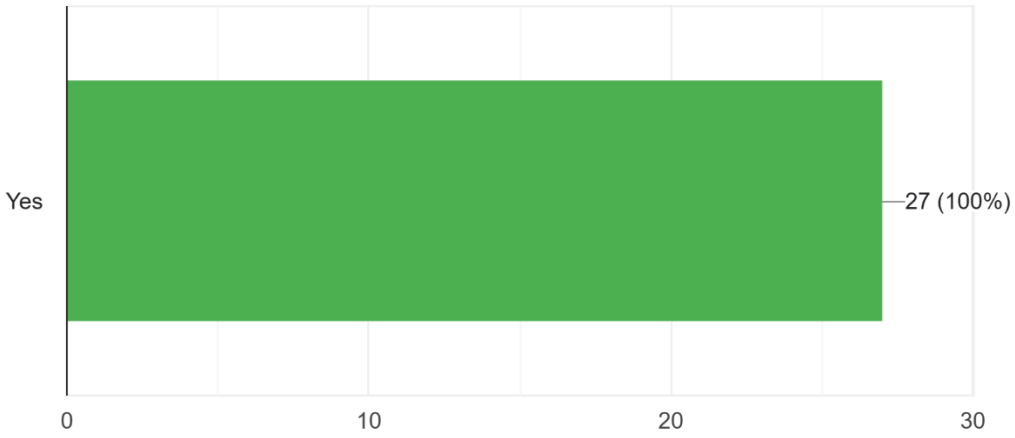
3. **The New Analog: Listening and Reconnecting in a Digital World** Krukowski, Damon. *The New Analog: Listening and Reconnecting in a Digital World*. The New Press, 2017.

17.Appendix

17.1 Qualitative survey form responses (self -reported) :

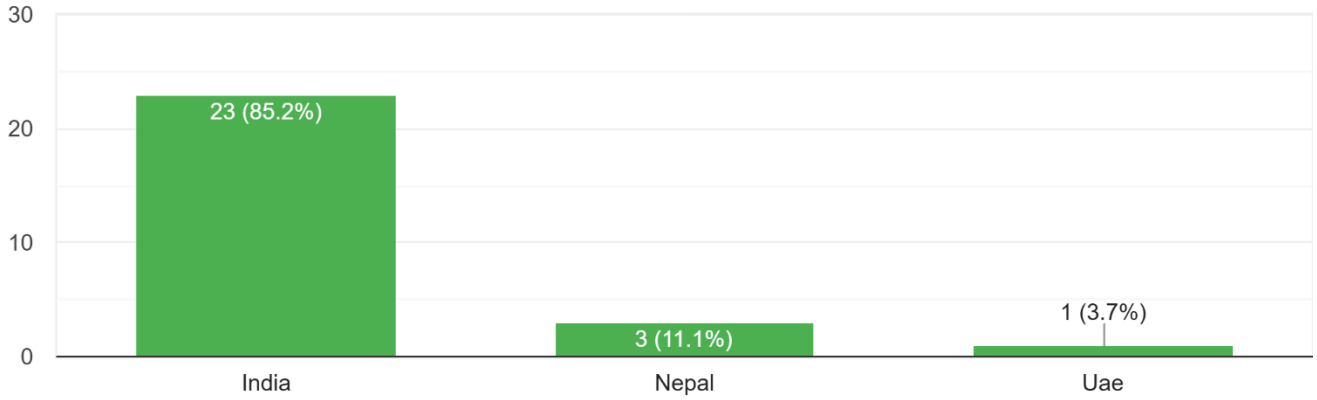
I understand the purpose of this study and agree to have my anonymized responses used for academic research

27 responses



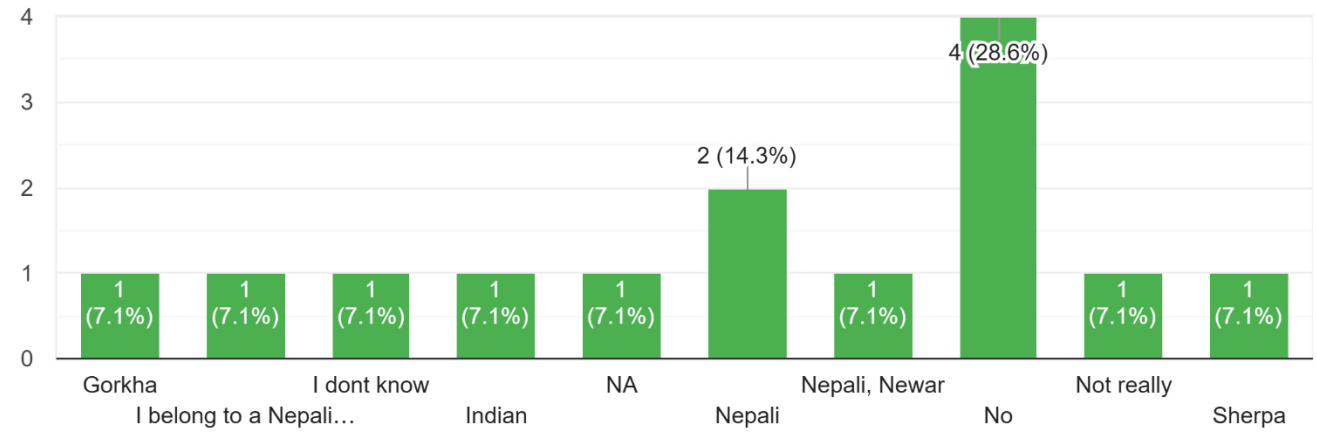
Which country do you currently live in?

27 responses



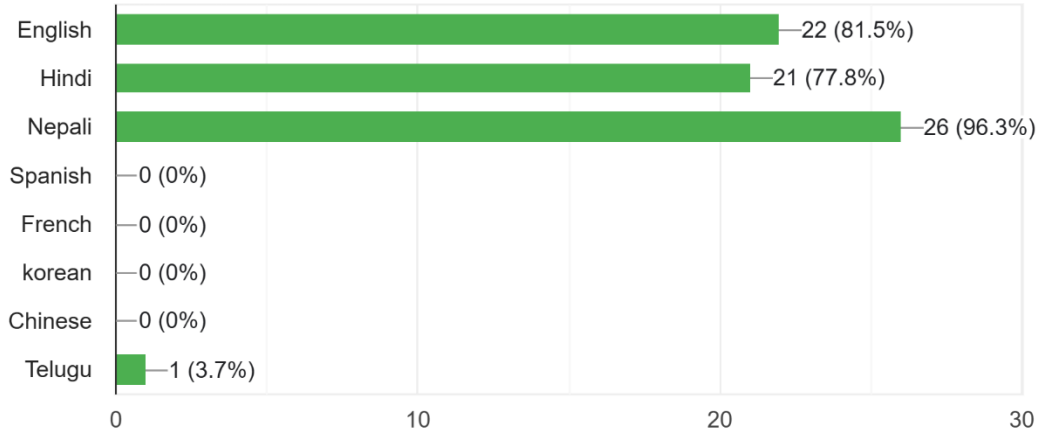
Do you identify with a specific cultural or ethnic background? If so, please specify.

14 responses



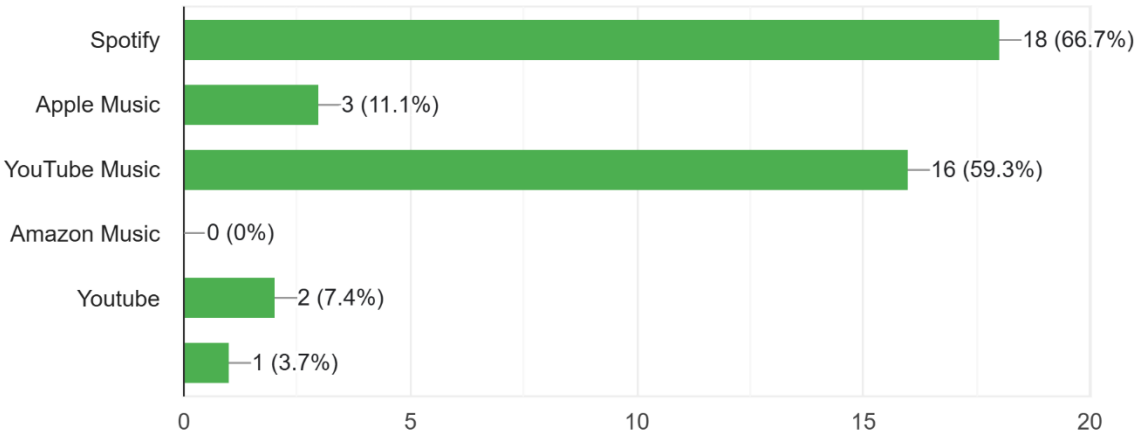
What language(s) do you speak fluently?

27 responses



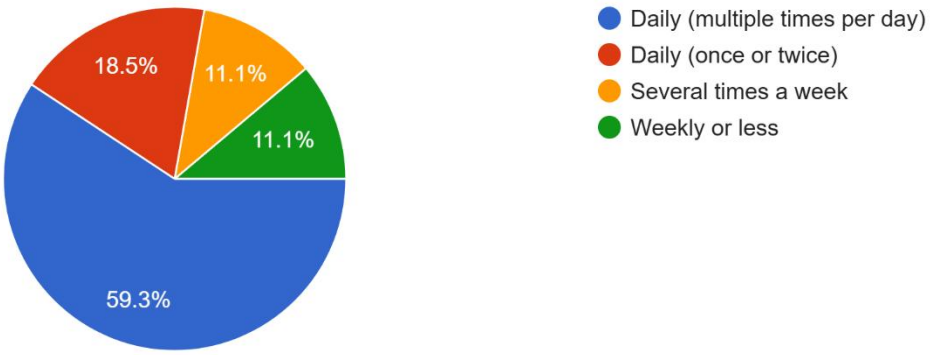
Which music streaming platforms do you use regularly ?

27 responses



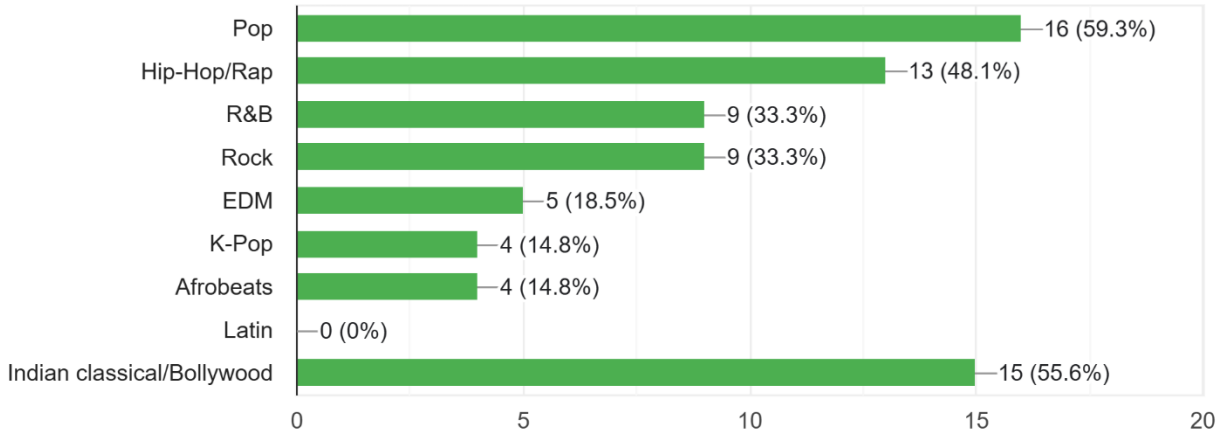
How often do you listen to music?

27 responses



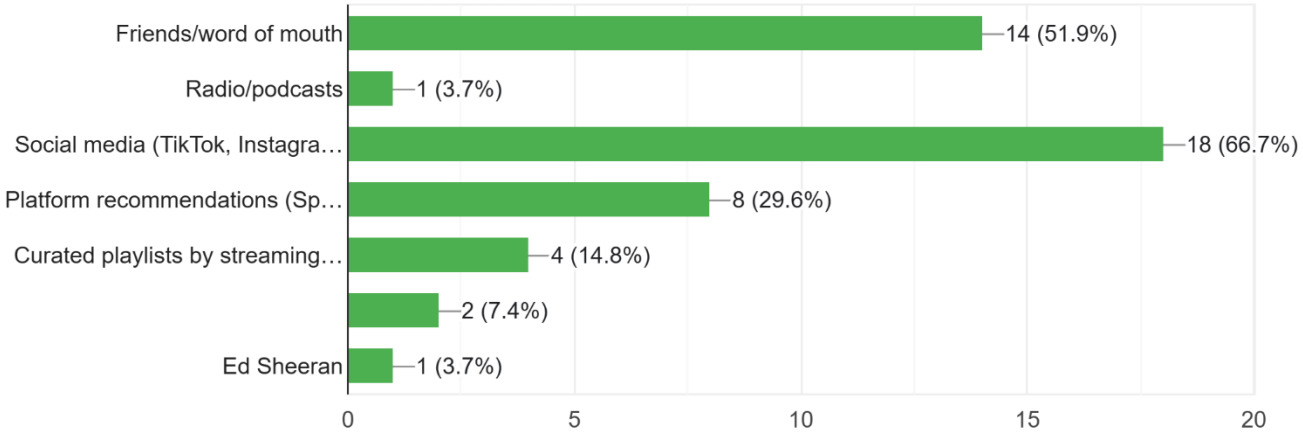
Which music genres do you often listen to? (Select all that apply.)

27 responses



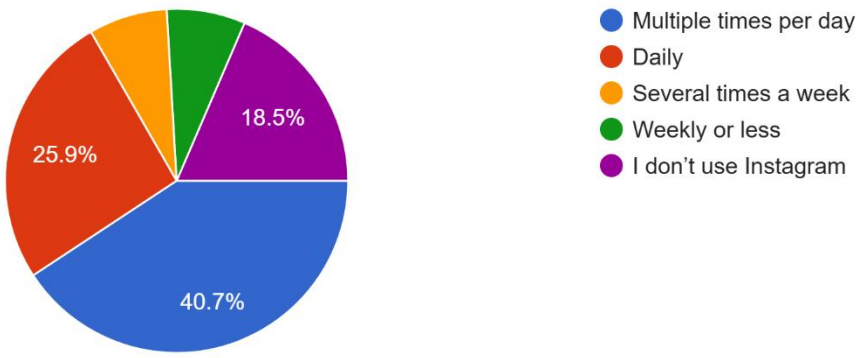
How do you usually discover new music? (Select all that apply.)

27 responses



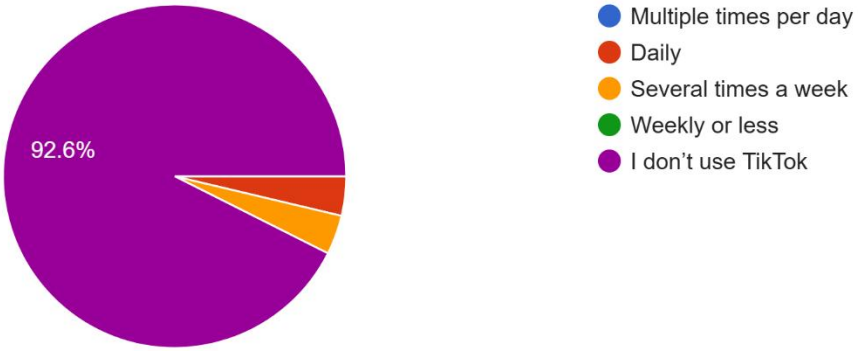
How often do you use Instagram?

27 responses



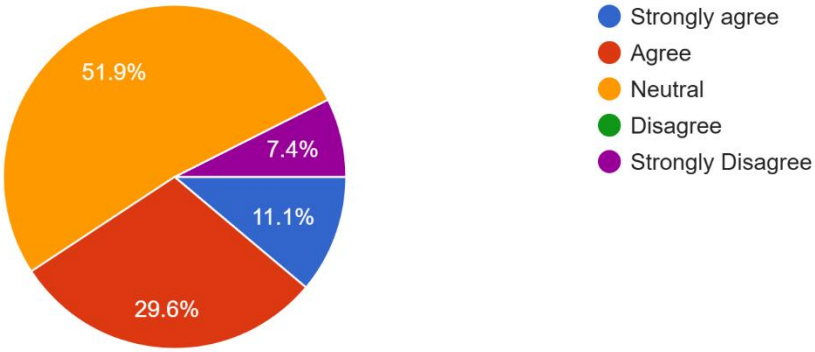
How often do you use TikTok?

27 responses



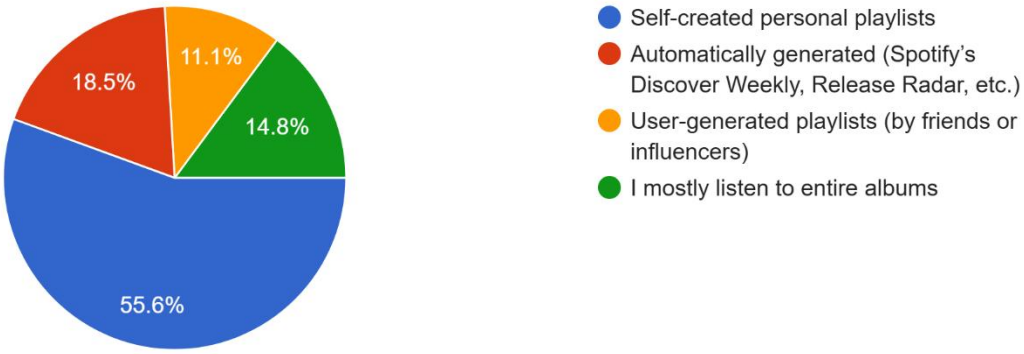
TikTok/Instagram trends significantly influence what music I listen to.

27 responses



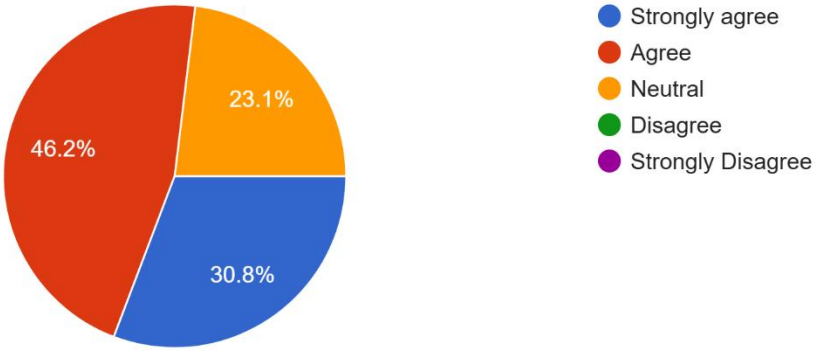
Which type of playlists do you listen to most often?

27 responses



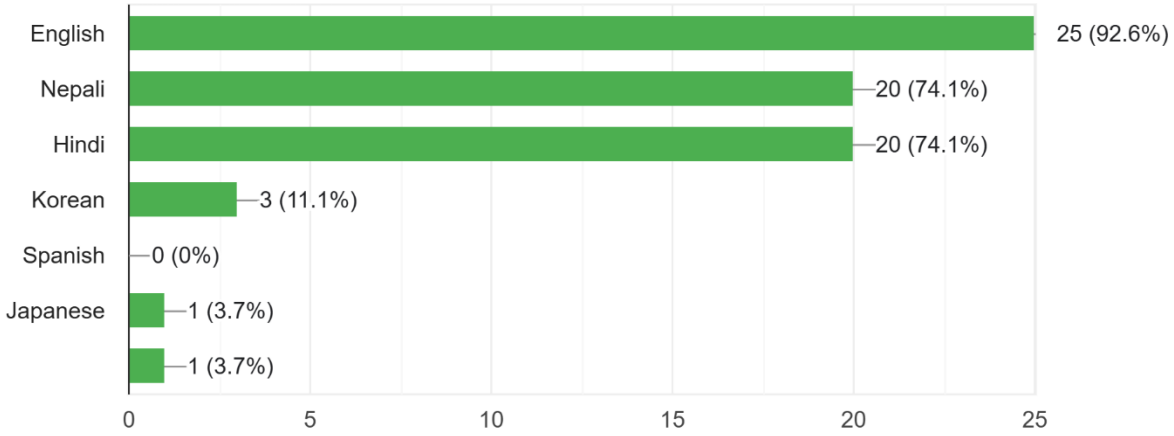
I actively seek out music from cultures or languages different from my own

26 responses



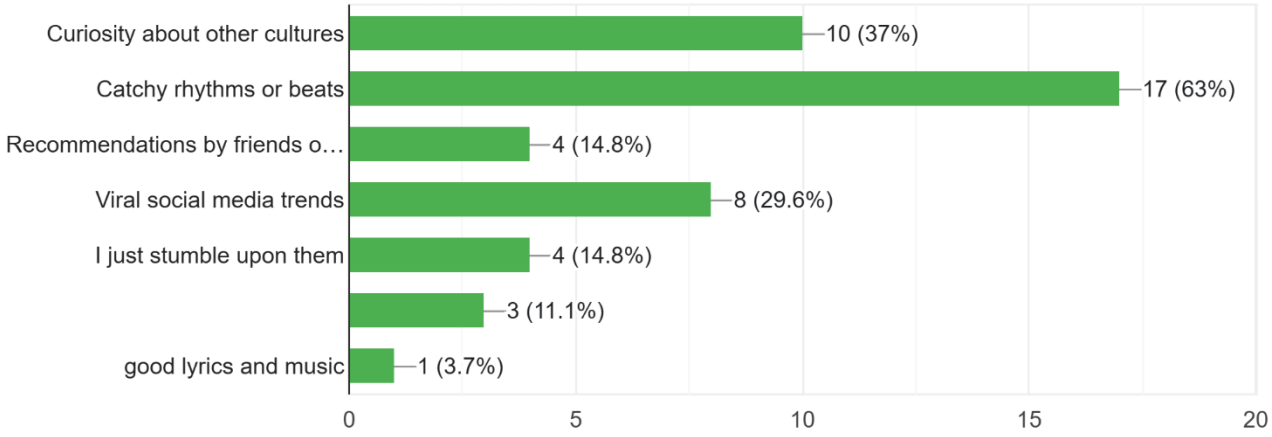
In which languages do you often listen to music?

27 responses



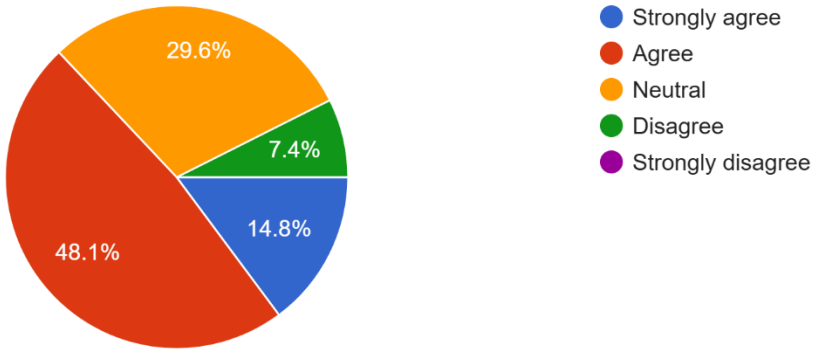
What motivates you to listen to music from other cultures? (Select all that apply)

27 responses



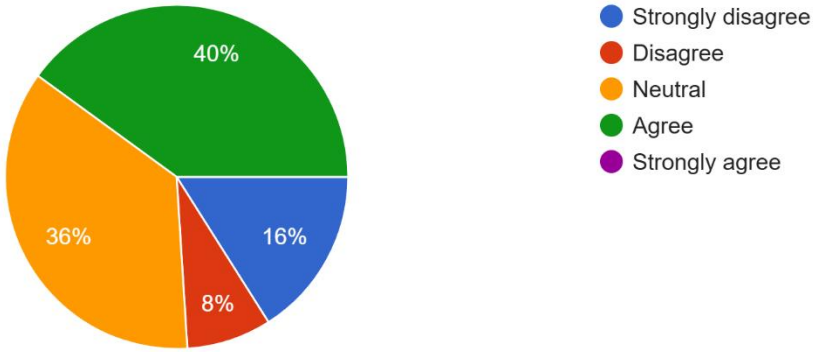
Streaming platform recommendations (e.g., Spotify's Discover Weekly) accurately reflect my music taste.

27 responses



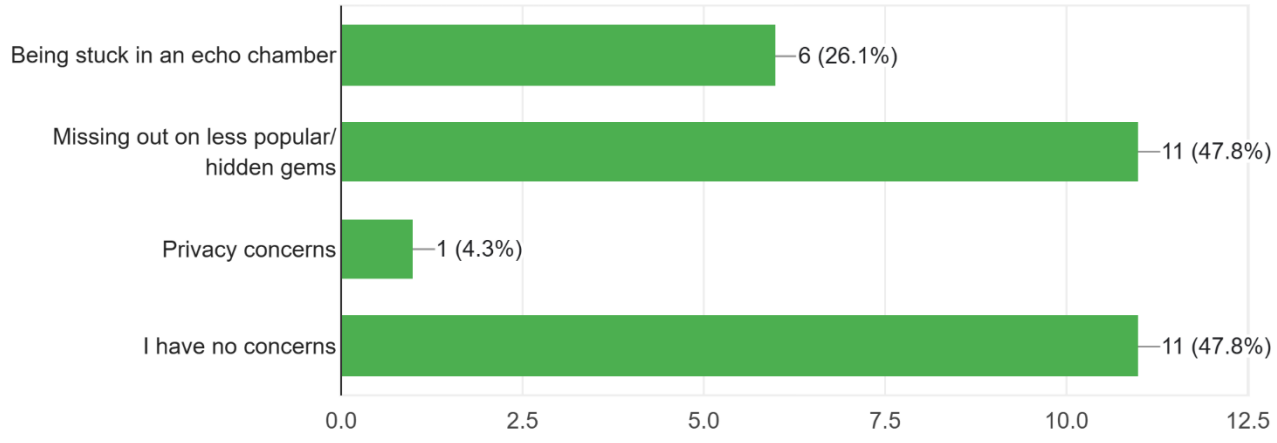
Overall, I am satisfied with how algorithms recommend music to me

25 responses



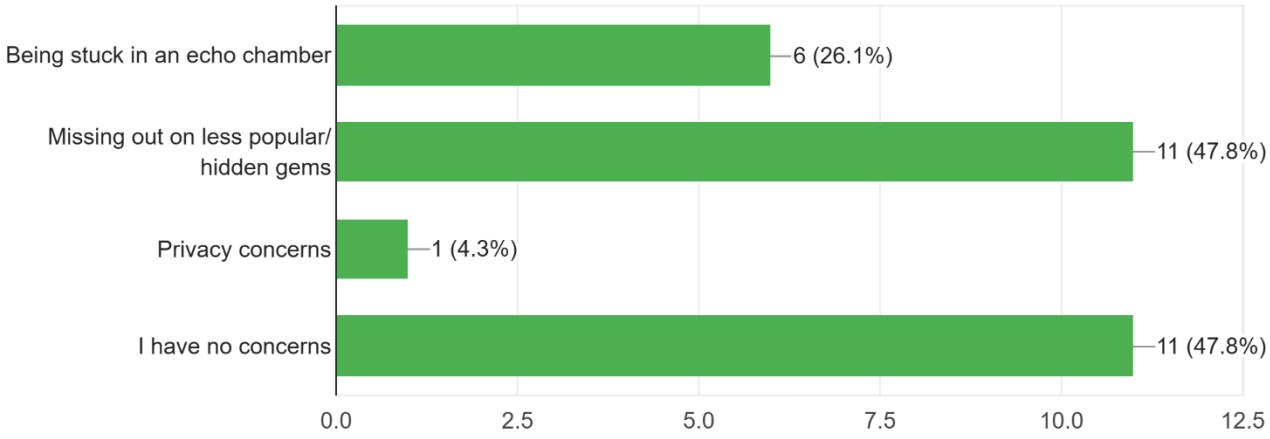
What concerns, if any, do you have about algorithmic recommendations? (Select all that apply.)

23 responses



What concerns, if any, do you have about algorithmic recommendations? (Select all that apply.)

23 responses



17.2 Meeting Logs:

MSc. Project Logbook Entry Sheet

Meeting Logbook Entry Sheet

Meeting No: 01 **Date: 03/02/2025**

Start Time: 2:00 PM **End Time: 2:45 PM**

Items Discussed:

1. *Dissertation Overview*
   - Presented the research introduction, core objectives, and key research questions.
2. *Data Collection Progress*
   - Discussed delays due to privacy concerns and the use of Google Forms for Spotify data and surveys.
   - Supervisor suggested using **QR codes** to distribute survey links more effectively within the college.
   - Updated on **TikTok Research API application**, which is still pending approval.
3. *Literature Review*
   - Confirmed that literature review is being conducted alongside data collection.
   - Discussed relevant research papers and sources.
   - Supervisor advised writing the review **independently** rather than relying on AI for better comprehension and academic development.
4. *Machine Learning Models*
   - Reviewed planned ML models for analysis.
   - Supervisor recommended **avoiding overly complex models**, as qualitative survey data also supports the research.

5. *Meeting Frequency*

- Agreed to hold **biweekly meetings** to ensure steady progress and structured discussions.

Achievements:

- Received guidance on increasing survey participation using QR codes.
- Confirmed that literature review and data collection should be done in parallel for efficiency.
- Gained valuable feedback on machine learning model selection—prioritizing simplicity over complexity.
- Established a biweekly meeting schedule to ensure steady progress.

Problems:

- Data collection delays due to privacy concerns.
- TikTok Research API approval still pending, which may affect data availability.
- Need to refine literature review approach while balancing dissertation workload.

Tasks for Next Meeting:

- Work on expanding survey reach within the college using QR codes.
- Continue literature review and ensure it is self-written for better understanding.
- Refine machine learning model selection, ensuring a balance between interpretability and accuracy.
- Follow up on TikTok API approval and explore alternatives if needed.



Student Signature



Supervisor Signature

MSc. Project Logbook Entry Sheet

Meeting Logbook Entry Sheet

Meeting No: 02 **Date: 30/03/2025**

Start Time: 4:00 PM **End Time: 4:20 PM**

Items Discussed:

1. *Literature Review Feedback*
   - Discussed the initial literature review draft submitted to the supervisor.
   - Supervisor suggested making the review less descriptive and more critical, particularly in evaluating the methods and findings of the papers.
   - Recommended adding a summary table to clearly present the key contributions and gaps across reviewed studies.

2. *TikTok API Update*
   - Informed the supervisor that TikTok Research API approval is significantly delayed.
   - Decided to proceed without TikTok data and focus on Spotify API, Genius API, and user survey data for the analysis.

3. *Methodology Development and Data Work*
   - Informed that the methodology chapter is being developed in parallel with ongoing data collection.
   - Supervisor acknowledged the importance of starting early.

Achievements:

- Received constructive feedback on the literature review structure and content.
- Finalized decision to focus on available data sources (Spotify, Genius, and survey data) due to API access issues.
- Initiated writing of methodology section and began data extraction and preprocessing.

Problems:

- Continued delay in TikTok API approval, requiring a change in data scope.
- Need to revise literature review tone and format.

Tasks for Next Meeting:

- Revise the literature review with a more critical approach and include the recommended summary table.
- Continue writing the methodology and progress further with data analysis and model planning.
- Prepare an update on the preprocessing outcomes and initial data insights.



Student Signature



Supervisor Signature

MSc. Project Logbook Entry Sheet

Meeting Logbook Entry Sheet

Meeting No: 03 **Date: 13/04/2025**

Start Time: 3:30 PM **End Time: 4:15 PM**

Items Discussed:

- Decision to drop TikTok Research API integration due to ongoing approval delays and shifting priorities.
- Encountered Spotify Web API rate limits when extracting large track batches; explored request-throttling and caching strategies.
- Continued development of the methodology chapter: refined primary-genre assignment procedure and formalized Polyphonic Score computation.
- Reviewed alternative data sources: Genius API for lyrics metadata and the TopByCountries dataset for geographic analysis.
- Outlined preliminary EDA plan: key audio-feature distributions, country-level summaries, and correlation matrices

Achievements:

- Officially pivoted research scope away from TikTok API, consolidating on Spotify and Genius data.
- Documented Spotify API constraints and drafted a caching/throttling module to mitigate rate-limit errors.
- Completed first draft of methodology subsections on genre harmonization and Shannon-entropy–based diversity scoring.
- Produced an EDA outline specifying datasets, target variables, and visualization types.

Problems:

- Persistent Spotify API rate-limit errors slowing bulk data collection.
- Absence of TikTok data reduces insight into social-media–driven virality patterns.
- Uncertainty around Genius API rate limits and data completeness for lyrics context.

Tasks for Next Meeting:

- Implement and test Spotify API caching and rate-limit handling in Python scripts.
- Run initial data-cleaning scripts on TopByCountries and GlobalWeekly datasets; report on missing-value patterns.



Student Signature



Supervisor Signature

MSc. Project Logbook Entry Sheet

Meeting Logbook Entry Sheet

Meeting No: 04 **Date: 27/04/2025**

Start Time: 2:00 PM **End Time: 2:45 PM**

Items Discussed:

- Began EDA and defined visualization goals.
- Built HashMap for fast Spotify data lookups.
- Cleaned & merged core datasets.
- Generated genre co-occurrence prototype.

Achievements:

- Query time cut by ~80% via HashMap indexing.
- Unified maindata, GlobalWeekly, TopByCountries.
- Computed basic EDA metrics by genre/country.
- Created initial network edge list..

Problems:

- High memory usage on full datasets.
- Inconsistent country-code formats.
- Slow matplotlib plots with large data.

Tasks for Next Meeting:

- Optimize data ingestion (chunking, Parquet).
- Finalize network-analysis thresholds.
- Produce key EDA plots (histograms, bar charts).
- Draft EDA methodology write-up.



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Supervisor Signature

MSc. Project Logbook Entry Sheet

Meeting Logbook Entry Sheet

Meeting No: 05 **Date: 21/05/2025**

Start Time: 3:30 PM **End Time: 4:15 PM**

Items Discussed:

Reviewed completed EDA, network analysis, Polyphonic Scores and XGBoost genre-classifier.

Dropped NLP-sentiment component to focus scope.

Examined validation metrics (65 % accuracy; top features: popularity, acousticness, track_age).

Agreed that all objectives are met and project is complete.

Achievements:

- Finalized data-processing and feature-engineering pipelines.
- Validated XGBoost model across 32 genres with clear baseline performance.
- Completed visualization of genre networks and diversity scoring.
- Consolidated all methodological and analytical work into dissertation draft.

Problems:

- Remaining class-imbalance challenges in minority genres.
- Limited time prevented integration of lyrical-sentiment features.
- Early API rate-limit issues reduced dataset size.

Project Status:

– Project work is now finished. No further meetings or tasks pending



Student Signature



Supervisor Signature